CORE MATHEMATICS GRADE 10: TRIGONOMETRY 1

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.4: Trigonometry 1
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Trigonometric ratios of acute angles (sine, cosine, tangent) — definitions using right-angled triangles and calculation using mathematical tables and calculators – Sines and cosines of complementary angles — relationship between sin θ and cos(90° − θ) – Relationship between sine, cosine, and tangent of acute angles (tan θ = sin θ / cos θ) – Trigonometric ratios of special angles: 30°, 45°, 60°, and 90° — derived using an isosceles right-angled triangle and an equilateral triangle – Application of trigonometric ratios in solving right-angled triangles – Angles of elevation — definition, identification, and calculation in real-life contexts – Angles of depression — definition, identification, and calculation in real-life contexts – Real-life applications of trigonometry (surveying, construction, navigation, height and distance problems in Kenyan contexts)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the trigonometric ratios of acute angles from mathematical tables and calculators, b) relate sines and cosines of complementary angles, c) relate the sine, cosine, and tangent of acute angles, d) determine trigonometric ratios of special angles: 30°, 45°, 60° and 90° using triangles, e) apply trigonometric ratios to angles of elevation and depression, f) reflect on the use of trigonometry in real-life situations.
Core Competencies	– Communication and Collaboration: the learner works in groups to draw right-angled triangles, generate tables of trigonometric ratios for complementary angles, and discuss patterns and relationships observed — sharing findings with peers and the teacher. – Critical Thinking and Problem Solving: the learner analyses real-life height-and-distance problems (e.g., estimating the height of the Kenyatta International Convention Centre or a mobile phone mast in the Rift Valley) using trigonometric ratios, angles of elevation, and angles of depression. – Learning to Learn: the learner uses mathematical tables and calculators to systematically look up and verify trigonometric ratios, building independent research and self-correction habits. – Self-Efficacy: the learner gains confidence by deriving the exact values of trigonometric ratios for special angles (30°, 45°, 60°, 90°) from first principles using triangles, rather than memorising them.

Core Values	– Responsibility: the learner carefully uses geometrical sets, rulers, and calculators as shared learning resources, returning them in good condition and ensuring accurate constructions when drawing triangles used to derive trigonometric ratios. – Integrity: the learner records measured angles and computed ratios honestly, without copying or fabricating results, so that conclusions about complementary angles and special angles are trustworthy. – Respect: the learner appreciates each group member's contribution during discussions on real-life applications of trigonometry, listening attentively when peers present their methods for solving angles of elevation and depression problems.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners pose questions such as "How can we find the height of a tree or building without climbing it?" to frame the need for trigonometric ratios, directly motivating the unit's anchoring phenomenon. – Using Mathematics and Computational Thinking: learners use ratio calculations, tables, and calculators to determine sine, cosine, and tangent values and to solve for unknown sides and angles in right-angled triangles set in Kenyan real-life contexts. – Analysing and Interpreting Data: learners generate tables of trigonometric ratios for a range of acute angles, identify patterns (e.g., $\sin \theta = \cos(90° - \theta)$), and interpret what those patterns reveal about the structure of right-angled triangles. – Constructing Explanations and Designing Solutions: learners use the isosceles right-angled triangle and the equilateral triangle to construct exact explanations for why $\sin 30° = 0.5$, $\sin 45° = \sqrt{2}/2$, etc., and design step-by-step solution methods for elevation and depression problems. – Obtaining, Evaluating, and Communicating Information: learners present their group findings on special angles and real-life trigonometry applications to the class, evaluating each other's reasoning and communicating conclusions clearly.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development — Environmental Education: the learner identifies right-angled triangle shapes from the immediate environment (roof trusses of Kenyan homes, sloped hillsides in the Rift Valley, ramps) and uses them as authentic contexts for defining and applying trigonometric ratios. – Social Cohesion: the learner collaborates in culturally diverse groups to measure, calculate, and present findings on angles of elevation and depression, respecting different approaches and valuing teamwork. – Life Skills — Decision Making: the learner applies trigonometry to practical decision-making scenarios, such as determining the safe angle of a ladder leaning against a Nairobi building or estimating the height of a communications mast, reinforcing that mathematics supports sound real-world judgement. – Safety and Security: the learner discusses how engineers and surveyors use angles of elevation and depression in constructing safe roads, bridges, and buildings in Kenya, appreciating the role of trigonometry in public safety.
Career Connections	– Land Surveying and Geomatics: trigonometric ratios are the foundation of surveying — Kenyan land surveyors use angles of elevation and depression daily to measure distances, map terrain, and demarcate plots across counties. – Civil and Structural Engineering: engineers designing bridges over the Tana River, roads through the Aberdare Ranges, or buildings in Nairobi's CBD rely on trigonometry to calculate slopes, load angles, and structural dimensions. – Architecture: Kenyan architects use trigonometric ratios when designing roof pitches, ramps, and staircases to meet safety standards and aesthetic requirements. – Navigation and Aviation: pilots and maritime navigators operating out of Jomo Kenyatta International Airport or the Port of Mombasa use angles of elevation and depression together with trigonometric ratios for flight path calculations and docking manoeuvres. – Astronomy and Meteorology: scientists at the Kenya Meteorological Department use angular measurements and trigonometry to track weather balloons, satellite dishes, and solar panel inclinations. – Teaching and Education: mathematics teachers across Kenya's 47 counties require deep understanding of trigonometry to effectively deliver the CBC curriculum at Senior School level.
Focus for Lessons	This unit introduces learners to the foundational concepts of trigonometry through the lens of a compelling real-life anchoring phenomenon: estimating the height of tall structures and natural features without direct measurement. Over 8 lessons, learners

	progress from constructing and naming the trigonometric ratios of acute angles, through discovering the elegant relationships between complementary angles and special angles, to confidently applying sine, cosine, and tangent in solving angles of elevation and depression problems drawn from Kenyan contexts such as flag poles in school compounds, communication masts in rural counties, and the slopes of Mt. Kenya. The pedagogical approach emphasises hands-on triangle construction, collaborative group inquiry, use of mathematical tables and calculators, and sense-making discussions that connect abstract ratios to visible, puzzling real-world situations.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance — and how does this help Kenyans solve real problems in construction, surveying, and everyday life? KICD KEY INQUIRY QUESTIONS: 1. What is trigonometry? 2. How do we use trigonometry in real-life situations?
Anchoring Phenomenon	ANCHORING PHENOMENON — "How tall is that mast … and can we find out without climbing it?" At the start of the unit, the teacher projects or displays a photograph of a Kenya Power electricity transmission tower (or a mobile phone mast) standing near a school in Nakuru County. Students are told that the base of the mast is accessible, but the top is obviously unreachable. A student standing 40 m from the base looks up and estimates the angle to the top is roughly 50°. The class is asked: "Using only that horizontal distance and that angle, can we calculate how tall the mast is — without ladders, drones, or guessing?" This is genuinely puzzling because students at Grade 10 entry have strong algebra and geometry skills, yet they have no tool that links an angle to a ratio of sides. They know Pythagoras relates sides to sides, but here only one side and one angle are known. The phenomenon creates an authentic knowledge gap — a need for a new mathematical tool — that trigonometric ratios precisely fill. Throughout the unit, every new concept (sine, cosine, tangent; complementary angles; special angles; elevation and depression) is explicitly connected back to solving this and similar height-and-distance problems across Kenya.
Storyline Thread	Lesson 1 — Noticing the Phenomenon & Building the Need: Teacher presents the Kenya Power mast photograph and poses the driving question. Students attempt (and struggle) to find the mast height with only algebra — revealing the knowledge gap. Learners identify right-angled triangle shapes from the classroom and school environment (roof trusses, ramps, staircases). Groups draw right-angled triangles of different sizes with the same acute angle and measure the three sides, recording opposite, adjacent, and hypotenuse. They notice the ratios of pairs of sides stay constant for a fixed angle — sparking curiosity about what these ratios mean. Lesson 2 — Defining Sine, Cosine, and Tangent: Learners formally define $\sin \theta$ = opposite/hypotenuse, $\cos \theta$ = adjacent/hypotenuse, and $\tan \theta$ = opposite/adjacent using their triangles from Lesson 1. They practise labelling sides relative to different angles in the same triangle. Introduction of mathematical tables: learners use trigonometric tables to find sin, cos, and tan for given acute angles and back-calculate angles from given ratios. Calculators are also introduced as a checking tool. Groups calculate the sine, cosine, and tangent of several angles and record results in a class table on the board. Lesson 3 — Complementary Angles ($\sin \theta = \cos(90° - \theta)$): Learners generate a table of complementary angle pairs (e.g., 30° & 60°, 25° & 65°, 42° & 48°) and record their sines and cosines from mathematical tables. They notice and articulate the pattern: $\sin \theta = \cos(90° - \theta)$ and $\cos \theta = \sin(90° - \theta)$. Teacher guides a geometric proof using a single right-angled triangle viewed from both acute angles. Learners practise applying the relationship to simplify expressions and solve for unknown angles.

	Lesson 4 — Relationship Between sin, cos, and tan: Learners use the values in their table from Lesson 2 to compute $\sin \theta / \cos \theta$ for multiple angles and compare the result with the recorded $\tan \theta$ values — discovering $\tan \theta = \sin \theta / \cos \theta$. Teacher formalises the identity. Learners practise using the identity to find a missing ratio when two are known and to verify calculator outputs. Short class activity: given $\sin \theta = 3/5$, find $\cos \theta$ and $\tan \theta$ without a calculator. Lesson 5 — Trigonometric Ratios of Special Angles (45° and 30°/60°): Learners construct an isosceles right-angled triangle with legs of 1 unit to derive $\sin 45° = \cos 45° = 1/\sqrt{2}$ and $\tan 45° = 1$. They then construct an equilateral triangle with side 2 units, bisect it to form two 30°-60°-90° triangles, and derive $\sin 30° = \cos 60° = 1/2$, $\sin 60° = \cos 30° = \sqrt{3}/2$, $\tan 30° = 1/\sqrt{3}$, and $\tan 60° = \sqrt{3}$. For 90° and 0°, learners use limit arguments with their tables. Groups compile a special-angles reference card in Kenyan flag colours. Lesson 6 — Solving Right-Angled Triangles Using Trigonometric Ratios: Learners apply all three ratios (and the special angles) to solve for unknown sides and angles in right-angled triangles set in Kenyan contexts — e.g., a ramp at a Kisumu hospital, a sloped road in Kericho tea plantation, a telecommunications tower in Garissa. Emphasis on choosing the correct ratio (SOH-CAH-TOA), showing clear working, and interpreting answers in context. Pairs check each other's solutions and discuss errors. Lesson 7 — Angles of Elevation and Depression: Teacher uses the mast photograph (anchoring phenomenon) and the Mombasa cliff supporting phenomenon. Learners define angle of elevation (measured upward from horizontal) and angle of depression (measured downward from horizontal) with diagrams. Step-by-step method for setting up and solving elevation/depression problems is developed collaboratively. Learners solve a set of graded problems: flag pole shadow in Nairobi, height of a building seen from across a Kisumu street, distance of a boat from the Fort Jesus cliff. Groups present solutions explaining each step. Lesson 8 — Synthesis, Real-Life Applications, and DQB Review: Learners revisit the anchoring phenomenon and calculate the height of the Kenya Power mast (40 m base, 50° elevation angle) — experiencing the satisfaction of solving the original puzzle. Class updates the Driving Question Board, identifying which questions have been answered and which open questions remain. Learners reflect in writing on one real-life profession in Kenya that uses trigonometry and how. Assessment activity: each learner independently solves a multi-step problem involving both an angle of elevation and an angle of depression, set on the shores of Lake Victoria.
Supporting Phenomena	– The leaning roof of a Kenyan mabati house: learners observe a corrugated-iron roof that slopes at an angle and ask "if I know the horizontal span and the roof angle, can I calculate how long the roofing sheet must be?" — motivating the use of cosine and tangent in right-angled triangles. – The shadow of a flag pole at a Nairobi school: at a particular time of day, a vertical flag pole casts a measurable shadow on the ground; learners ask "the shadow is 6 m long and the sun's elevation angle is 40° — how tall is the flag pole?" — anchoring the angle-of-elevation concept. – A boat observed from a cliff at Mombasa's Fort Jesus: learners view an image of a person on top of a cliff looking down at a dhow on the Indian Ocean and ask "if the cliff is 30 m high and the angle of depression to the boat is 25°, how far is the boat from the cliff base?" — anchoring the angle-of-depression concept. – The symmetrical peak of Mt. Kenya visible from Nanyuki: learners examine a photograph showing a climber on a flat plain looking up at the summit, given horizontal distance and elevation angle, and ask "can the mountain's height be estimated this way?" — connecting trigonometry to geography, science, and Kenyan national pride.

LESSON 1 (80 minutes): Noticing the Phenomenon & Building the Need: What Tool Is Missing?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and establish an authentic knowledge gap: students recognise that existing tools (algebra, Pythagoras) cannot find the height of a mast when only one side and one angle are known — creating a felt need for trigonometric ratios.
Knowledge	Students will know that a right-angled triangle has three sides (opposite, adjacent, hypotenuse) relative to a chosen acute angle, and that drawing similar right-angled triangles with the same acute angle produces constant ratios between pairs of sides.
Skills	Students will be able to: (a) identify right-angled triangles in the school environment; (b) draw right-angled triangles of different sizes sharing the same acute angle; (c) measure and label opposite, adjacent, and hypotenuse sides; (d) compute and record the three possible side-pair ratios and observe their constancy.
Attitudes	Students will appreciate that mathematics grows from real problems, value collaborative inquiry, and show curiosity and intellectual honesty when their existing knowledge is insufficient to solve the mast problem.
Key Inquiry Question	What is trigonometry, and why do we need a new mathematical tool when we already know algebra and Pythagoras?
Purpose in Storyline	This is the opening lesson of the unit. It presents the Kenya Power mast photograph, poses the driving question, exposes the knowledge gap through a productive struggle activity, grounds students in the language of right-angled triangles, and opens the Driving Question Board — establishing the investigative thread that every subsequent lesson will build on.
Safety Notes	Students will use rulers and protractors with sharp edges — remind learners to place instruments flat on the desk and pass them handle-first. If the lesson moves outdoors to observe roof trusses or ramps, students must remain on designated pathways and not climb any structure.

B. LESSON OVERVIEW
Lesson 1 launches the anchoring phenomenon for the entire Trigonometry 1 unit. The teacher displays a photograph of a Kenya Power electricity transmission tower near a school in Nakuru County and poses the driving question: 'Using only a horizontal distance of 40 m and an estimated angle of 50°, can we calculate the mast's height without ladders, drones, or guessing?' Students first attempt — and visibly struggle — to solve this with algebra alone, exposing the knowledge gap. They then identify right-angled triangle shapes across the classroom and school compound (roof trusses, staircase risers, door-frame diagonals, wheelchair ramps). Working in groups of four, learners draw multiple right-angled triangles of different sizes all sharing the same acute angle (e.g. 40°), measure opposite, adjacent, and hypotenuse sides with rulers, compute the three side-pair ratios, and record findings in a shared table. The lesson closes with groups posting their most burning questions on the inaugural Driving Question Board and constructing an initial pencil sketch model of the mast scenario, annotating what they know and what they still need.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
1 — PREDICT 📹 **VIDEO:** Trigonometry	Students view the Kenya Power mast photograph projected at the front. They read the scenario card	1. Project the mast photograph and read the scenario aloud slowly. Say exactly: 'I want every	Productive Struggle Elicitation — students attempt an unsolvable problem with current tools. The	Observable indicator: Do students correctly set up a right-angled triangle diagram with the 40 m

Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	on their desks: 'A student in Nakuru stands 40 m from the base of a transmission tower and estimates the angle to the top is about 50°. No ladder. No drone. Only a tape measure and a protractor. Can you find the height?' Working individually for 4 minutes, learners write their predicted height (with working) in their exercise books, then share with a shoulder partner. Groups of four then agree on one 'best attempt' using only algebra and Pythagoras and present it on a mini-whiteboard or piece of manila paper. The teacher circulates silently, reading attempts without correcting.	one of you to try — use every maths tool you already own. You have four minutes. Go.' Set a visible timer. Allow genuine struggle — do NOT hint. After 4 minutes say: 'Share your attempt with your partner. You have 90 seconds.' After partner share, cold-call THREE groups (select one confident group, one who wrote 'impossible', one who guessed). For each, ask: 'Walk us through your reasoning.' After all three, pose to the whole class: 'Hands up — who feels they got a definite, justified answer?' (Expect very few or no hands.) Say: 'Good. That discomfort you feel is exactly where mathematics grows. Today we are going to find out what tool is missing — and why it matters for Kenyans who build towers, bridges, and buildings every single day.'	strategy surfaces prior knowledge (algebra, Pythagoras), makes the knowledge gap visceral and personally felt, and generates genuine motivation to learn the new tool. The shared mini-whiteboard display lets the class see that nobody — not even the strongest algebraist — can solve the problem yet, normalising the gap.	base and 50° angle labelled, even if they cannot solve it? Teacher notes which groups draw a triangle (shows geometric intuition) versus which write purely algebraic equations (shows over-reliance on algebra). Red flag: any student who claims to have solved it correctly — probe immediately with 'How did you find the second side without knowing the height?'
2 — OBSERVE 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Environment Scan (10 min): Groups of four receive a 'Triangle Hunt' recording sheet. They have 8 minutes to identify and sketch at least four real right-angled triangles visible from where they sit or stand (roof truss outline through the window, staircase profile, door-frame diagonal, ramp edge, set-square on the teacher's desk, corner of a textbook). For each, they label which angle they are 'standing at' and name the three sides as opposite (O), adjacent (A), and hypotenuse (H). Groups then share one finding each in a 2-minute gallery walk past each other's desks. Part B — Ratio Discovery Activity (25 min): Each group receives a protractor, ruler, and a blank A3	Part A: Say 'You have eight minutes — find four right-angled triangles in this room or visible through the window. Every triangle must be a real object, not one you drew. Sketch it, mark the right angle, and label O, A, H from one acute angle. Go.' Circulate, spend no more than 45 seconds per group. Prompt stuck groups: 'Look up — what shape is that roof truss making against the sky?' After 8 minutes, give a 30-second gallery walk signal. Part B: Distribute materials. Demonstrate ONE triangle on the board — draw a right-angled triangle with a 40° angle, measure sides, compute O÷H only — then STOP and say: 'Your job is to complete the other two triangles	Data-Pattern Recognition through Structured Measurement — learners generate their own empirical data rather than receiving a formula. This mirrors how historical mathematicians (including Islamic scholars who built on Greek work, later used by Kenyan surveyors) first discovered these relationships. By computing ratios themselves, students own the discovery. The three-triangle design controls for size variation, making the constancy of ratios unmistakable and surprising.	Observable indicator: Each group's O÷H values for their three triangles should cluster within ±0.03 of each other (measurement error). Teacher scans tables — any group with wildly varying ratios likely has angle measurement error; redirect them to re-check the 40° angle with the protractor. Listen for the word 'same' or 'constant' in group conversations — this signals the key insight is emerging. Collect one recording sheet per group at the end of this phase.

	sheet. They draw THREE right-angled triangles of different sizes, all with the same acute angle of 40° (marked and verified with the protractor). Side lengths must differ — e.g. hypotenuses of roughly 6 cm, 9 cm, and 12 cm. For each triangle they measure O, A, and H to the nearest mm and record in a group table with columns: Triangle, O (cm), A (cm), H (cm), O÷H, A÷H, O÷A. Groups compute all six ratio cells and record to two decimal places. They circle any pattern they notice.	and all three ratios. I want to see if you find what I think you will find.' Wait 15 seconds before releasing groups to work. Circulate every 3–4 minutes. Ask probing questions: 'Your O÷H for Triangle 1 is 0.64 and for Triangle 2 it is 0.65 — is that the same or different? What do you think is causing that tiny difference?' Cold-call TWO groups mid-activity to read their O÷H values aloud so the class hears emerging convergence. With 3 minutes remaining say: 'Write one sentence describing the pattern you see in your ratio columns.'		
3 — EXPLAIN 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups share their ratio tables. The teacher builds a class master table on the board, collecting each group's O÷H, A÷H, and O÷A values. Learners observe that — despite different triangle sizes — the ratios for the same 40° angle are nearly identical across all groups. The teacher formally names these ratios: sine (O÷H), cosine (A÷H), and tangent (O÷A), writing the names beside the class table. Students are then asked to return to the mast problem: 'If we know the angle is 50° and the adjacent side is 40 m, which ratio connects the opposite side — the height — to what we know?' Learners work in pairs to write the equation height ÷ 40 = [the ratio for 50°] and recognise they still need the actual value of that ratio — which they do not yet have. This re-anchors the knowledge gap: we need a way to find the ratio for any angle, not just 40°.	Build the class master table by cold-calling FOUR groups to read their three ratio values aloud. As you write them, say: 'Class — what do you notice about this column?' Wait 12 seconds. Accept responses. Then say: 'These ratios have names that mathematicians have used for over a thousand years — names that Kenyan engineers, surveyors, and builders use today. The first is called sine.' Write 'sine θ = Opposite ÷ Hypotenuse' on the board in large text. Repeat for cosine and tangent. Then return to the mast diagram and say: 'In our mast problem, the 40 m is the side next to the 50° angle — that is the adjacent side. The height is the side across from the 50° angle — that is the opposite side. So which ratio do we need?' Wait 10 seconds, then cold-call ONE student. After correct identification of tangent, say: 'Write this equation in your book: height ÷ 40 = tan 50°. Now — what is tan 50°?' Pause 10 seconds. Let students look puzzled. Say: 'That is our job for the next lesson. Today, you	Anchored Naming and Return to Phenomenon — new vocabulary (sine, cosine, tangent) is introduced only after students have independently discovered the ratios empirically, so the names feel like labels for something real rather than arbitrary definitions. Immediately re-applying the names to the mast problem closes the explain loop and re-opens the productive knowledge gap: we need the actual ratio value for 50°.	Observable indicator: Can each student correctly label O, A, H on the mast diagram relative to the 50° angle and write 'height ÷ 40 = tan 50°' unprompted? Teacher does a show-your-book check — walk the rows and tap the desk of any student whose equation is missing or incorrect. Provide a 30-second peer correction before moving on.

		have discovered that the ratio exists and that it is constant. That is enormous.'		
4 — DRIVING QUESTION BOARD (DQB) Creation 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner receives two sticky notes (or small cut pieces of paper if sticky notes are unavailable). On the FIRST note they write one question they genuinely want answered about the mast problem or about the ratios they just discovered. On the SECOND note they write one connection: somewhere in Kenya — beyond masts — where knowing a height or distance without measuring directly would be useful. Groups share their notes aloud (30 seconds per group), then stick them on the class DQB (a dedicated section of the board or a sheet of manila paper pinned to the wall), organised by the teacher into three emerging columns: 'What are the ratios for other angles?', 'How do we use ratios to solve real problems?', and 'Where is this used in Kenya?'. The teacher reads the board aloud and says: 'Every lesson in this unit will answer at least one of these questions. Watch this board grow.'	Distribute sticky notes. Say: 'You have three minutes — one genuine question, one Kenyan connection. These will live on our board for the whole unit.' After writing, cold-call FIVE groups to read their question note before sticking it up. As you organise the board, narrate: 'I am grouping Amina's question with Juma's — both are asking about how we get the ratio value for any angle. That is exactly Lesson 2.' Point to a connection note that mentions building construction and say: 'Who wrote this? Tell us in one sentence.' After the student speaks, say: 'Keep that in mind — we will come back to construction in Lesson 5.' Take a photograph of the DQB (or ask a student to sketch it) to preserve it.	Question Storming on a Driving Question Board — making student questions visible and permanent shifts epistemic authority toward learners and creates a public record of the unit's intellectual journey. Organising notes into thematic columns shows students that their individual curiosity maps onto the curriculum structure, building metacognitive awareness of their own learning trajectory.	Observable indicator: Are student questions specific and phenomenon-linked (e.g. 'How do we find tan 50° without measuring?') rather than generic ('What is maths?')? Questions that are too vague signal the student has not yet connected to the phenomenon — pull them aside and ask: 'What confused you most about the mast problem? Write that as your question.'
5 — MODEL BUILDING 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 [Search ARES	Each student draws an initial pencil model in their exercise book — a labelled sketch of the mast scenario. The model must include: the mast (vertical line), the ground distance of 40 m (horizontal line), the 50° angle at the student's standing position, and labels for the right angle at the base of the mast. Students label the three sides O (height — unknown), A (40 m — known), and H (slant distance — unknown). They annotate two things in a different colour: (1) what they now KNOW	Say: 'Close your eyes for five seconds and picture the mast in Nakuru. Open. Now draw it — exactly as it is, with everything we know and everything we don't. You have five minutes.' Circulate and check that every student has the right angle marked at the base (not at the student's feet). Correct this quietly: 'Where is the right angle in a right-angled triangle? Is the mast leaning, or is it straight up from the ground?' After 5 minutes say: 'Annotate your model — green pen or pencil for what we know, red for	Initial Annotated Model Construction — asking students to draw and annotate a model externalises their mental representation of the phenomenon and makes knowledge-gaps visually explicit. The two-colour annotation (known vs. needed) is a metacognitive scaffold that students will reuse in every subsequent lesson as the model becomes progressively more complete, creating a tangible record of unit-level learning growth.	Observable indicator: Does the student's model correctly place the right angle at the base of the mast (not at the observer's feet or at the top)? Is the 40 m labelled on the horizontal (adjacent) side and the unknown height on the vertical (opposite) side? Misplaced right angles are the most common early error — address with whole-class feedback at the start of Lesson 2 if more than 20% of books show this error.

for similar readings	— 'the ratio O÷A = tan 50° is constant for 50°'; (2) what they still NEED — 'the actual decimal value of tan 50°'. Below the diagram they write one sentence: 'I think we will find tan 50° by ___.' (open prediction). Groups share predictions with a neighbour in 60 seconds.	what we still need.' After annotation, cold-call TWO students to read their prediction sentence aloud. Do not confirm or deny either prediction — say: 'Both of those are worth investigating. Hold that thought until Lesson 2.' Collect books or ask students to keep the model visible at the top of tomorrow's page.		

D. TEACHER REFLECTION
After the lesson, consider: (1) Did the productive struggle in the Predict phase feel genuine — did students truly attempt the problem, or did they disengage quickly? If disengagement was high, the mast context may need to be made more personal (e.g. ask 'Has anyone seen this tower on the Nakuru–Nairobi highway?'). (2) In the ratio discovery activity, how close were each group's three ratio values to each other? If spread was large (>0.05), protractor use needs explicit re-teaching at the start of Lesson 2. (3) Did the DQB generate phenomenon-specific questions? If questions were generic, revisit the mast photograph at the start of Lesson 2 and ask students to re-read their question and sharpen it. (4) Were all learners — including those who are quieter or who struggle with measurement — meaningfully included in the group activity? Consider assigning explicit roles (Measurer, Recorder, Calculator, Reporter) in Lesson 2 to distribute participation. (5) Note the names of students who correctly identified tangent as the relevant ratio in the Explain phase — they are ready for extension tasks in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that right-angled triangles appear throughout the school environment (roof trusses, staircases, ramps) and that when multiple right-angled triangles share the same acute angle but differ in size, the ratio of any two chosen sides remains nearly constant across all triangles.
What did I learn?	Students learned that a right-angled triangle has three named sides — opposite, adjacent, and hypotenuse — defined relative to a chosen acute angle, and that the three constant side-pair ratios for a given angle are called sine (O÷H), cosine (A÷H), and tangent (O÷A).
How does this explain the phenomenon?	Students explained that to find the height of the Kenya Power mast in Nakuru, they need the equation height ÷ 40 = tan 50°, but they cannot yet solve it because they do not know the decimal value of tan 50° — identifying the precise knowledge gap that the rest of the unit will fill.

LESSON 2 (40 minutes): Defining Sine, Cosine, and Tangent — Naming the Ratios That Unlock the Mast	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners formally define sin θ, cos θ, and tan θ as ratios of named sides, then use mathematical tables and calculators to evaluate these ratios for acute angles — building the precise tool needed to calculate the height of the Nakuru mast.
Knowledge	– Define sine, cosine, and tangent of an acute angle as opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent respectively. – Identify and correctly label the opposite, adjacent, and hypotenuse sides relative to any chosen acute angle in a right-angled triangle. – Use mathematical tables and a calculator to find sin θ, cos θ, and tan θ for given acute angles, and to back-calculate an angle from a given ratio.
Skills	– Accurately label triangle sides (opposite, adjacent, hypotenuse) relative to different reference angles in the same triangle. – Read trigonometric tables fluently and cross-check results using a scientific calculator.
Attitudes	– Appreciate that precise mathematical definitions create powerful, universally applicable tools. – Show curiosity and persistence when navigating mathematical tables for the first time.
Key Inquiry Question	What is trigonometry — and how do we define and calculate the sine, cosine, and tangent of an acute angle?
Purpose in Storyline	Lesson 1 revealed that a constant side-ratio exists for every angle; Lesson 2 gives those three ratios their official names (sin, cos, tan), equips learners to evaluate them numerically using tables and calculators, and connects the resulting numbers directly to the mast problem: sin 50° ≈ 0.766 is the ratio that will eventually yield the mast's height.
Safety Notes	No specific hazards. Remind learners to handle mathematical tables and calculators with care and to avoid pressing random keys that could clear stored values.

B. LESSON OVERVIEW

This lesson is the definitional heart of the entire Trigonometry 1 sub-strand. Coming off Lesson 1 — where learners discovered empirically that the ratio of any two sides of a right-angled triangle depends only on the angle, not on the triangle's size — learners are now ready to receive the formal names: sine, cosine, and tangent. The lesson opens by revisiting the Nakuru mast photograph and the single incomplete calculation left on the board: "height = 40 m × ???". The "???" is the hook; by the end of the lesson learners will know it equals tan 50°, will look it up in a table, and will see the mast's height emerge as a real number for the first time.\n\nThe Observe Phase is the lesson's practical core. In pairs, learners draw two different right-angled triangles (angles of their choice between 20° and 70°), measure sides, compute the three ratios for both acute angles in each triangle, and record results in a shared class table on the board. They immediately notice that switching the reference angle swaps the "opposite" and "adjacent" labels — a crucial insight for avoiding the most common labelling error. Learners then open mathematical tables (sine, cosine, tangent pages) and locate the values for their measured angles, comparing table values to their computed ratios and experiencing the power of the table as a pre-computed library of ratios.\n\nIn the Explain Phase learners apply the formal definitions to the mast scenario: with θ = 50° and adjacent side = 40 m, they identify which ratio links angle to height (tangent), look up tan 50° ≈ 1.192, and calculate height ≈ 47.7 m. This is the first time the driving question yields a real numerical answer, and the moment is celebrated deliberately. The lesson closes with learners updating their Driving Question Board and revising their triangle model to include ratio notation alongside side labels.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Intro to the Pythagorean theorem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher re-displays the Nakuru mast photograph and writes on the board: 'A student stands 40 m from the base of the mast and measures the angle to the top as 50°. Which side-ratio connects this angle to the mast's height?' Each learner writes a prediction in their exercise book: name the ratio (even if uncertain), write a formula, and estimate the numerical value of that ratio. Learners then share with a shoulder partner for 90 seconds before the teacher cold-calls three students.	Display the mast image and say: 'In Lesson 1 we named the sides — opposite, adjacent, hypotenuse. Today we give the ratios their official names. But first — without opening your tables — predict: which ratio do you think links the 50° angle to the height of the mast? Write the ratio name if you know it, write the formula, and guess its value. You have 2 minutes — go.' [Allow 2 minutes of individual silent writing.] After 90 seconds of pair sharing, cold-call exactly 3 students: 'Achieng — what did you write?', 'Mutua — agree or disagree with Achieng, and why?', 'Wanjiku — what numerical value did you estimate?' [WAIT TIME: 12 seconds after each question before accepting an answer.] Record all three predictions on the board under the heading 'Our predictions — we will check these.' Do NOT confirm or correct yet. Say: 'Keep these predictions in mind — by the end of class we will know exactly who was closest.'	Prediction Anchoring — by committing a prediction to paper before instruction, learners activate prior knowledge from Lesson 1 and create a personal stake in verifying the answer. Recording predictions publicly on the board makes the knowledge gap visible and motivates the Observe Phase.	Scan written predictions as you circulate during the 2-minute writing window. Look for: (1) correct identification of 'opposite/adjacent' or 'tangent' — signals strong carry-over from Lesson 1; (2) confusion between opposite and hypotenuse — signals the labelling work in the Observe Phase is critical; (3) blank pages — identify these learners for targeted support during pair work.
Observe Phase 📹 **VIDEO:** Intro to the Pythagorean theorem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook)	In pairs, learners draw two fresh right-angled triangles in their exercise books — one with an estimated acute angle of about 35° and one with about 60° — using ruler and protractor. They measure all three sides to the nearest millimetre. For each triangle, they choose each acute angle in turn as the reference angle, label O, A, H correctly, and compute all three ratios (O/H, A/H, O/A) to 3 decimal places. They record all six sets of ratios in a personal table. Each pair then contributes one ratio value to a class table drawn on the	Distribute rulers and protractors. Say: 'Draw two right-angled triangles — make them different sizes. Estimate one angle at about 35° and in the other about 60°. Measure all three sides. Then, for EACH acute angle in EACH triangle, label which side is opposite, which is adjacent, which is hypotenuse — remember, these labels CHANGE when you switch which angle you look from. Compute O÷H, A÷H, O÷A. Record everything.' [Circulate for 8 minutes. At each pair, ask: 'You have switched to the other acute	Data-Gathering and Pattern Noticing — learners generate their own data (measured ratios), contribute to a collective class dataset, then compare against the authoritative mathematical table. This two-layer evidence structure (personal measurement → table confirmation) builds trust in the table as a reliable tool and surfaces the pattern that switching reference angles swaps O and A labels.	While circulating, check that every pair has correctly re-labelled sides when switching reference angles — this is the single most important conceptual move in the phase. If a pair has not changed labels, stop and ask: 'When you look from THIS angle, point to the side directly across from it. What do we call that side?' Look at the class table on the board: if the sin and cos columns for the same angle do not roughly sum near 1 (not exactly — that is Lesson 3), a labelling or calculation error likely exists. Identify and address

Source: CK-12 🔍 Search ARES for similar readings	board (columns: angle, sin, cos, tan; rows filled in by different pairs). Finally, learners open their mathematical tables to the sine, cosine, and tangent pages and locate the values matching their measured angles, comparing table values to their computed ratios and noting the difference (which should be small if measurements were accurate).	angle — have your labels changed? Show me.'] After 8 minutes, draw the class table on the board. Point to pairs one by one: 'Otieno and Kamau — give me your sin of 35°.' Fill in the table. Once 6–8 rows are filled say: 'Now open your mathematical tables to the sine pages. Find sin 35°. What do you see?' [WAIT TIME: 10 seconds.] Cold-call 2 students to read the table value aloud. Say: 'How close is the table value to your computed ratio? The small difference is measurement error — the table is exact.' Repeat for cosine and tangent pages. Then say: 'Notice that for the same angle, as sin goes up, cos goes down — we will explore why in Lesson 3. For now, simply observe it in the table.'		publicly.
Explain Phase 📹 **VIDEO:** Intro to the Pythagorean theorem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally defines the three ratios using the mnemonic SOH-CAH-TOA written on the board. Learners copy the three definitions into their books using both symbolic notation (sin θ = O/H) and word notation. The class then works through the mast problem together as a guided example: they identify the known side (adjacent = 40 m), the known angle (50°), and the unknown side (opposite = height). They choose tan θ because it links O and A without involving H. They look up tan 50° in the mathematical table (≈ 1.192), write the equation height = 40 × tan 50°, compute height ≈ 47.7 m, and check on a calculator. Learners then individually solve two practice problems: (a) a surveyor in Kisumu stands 25 m from a tree and measures the angle of elevation as 38° — find the tree's height using tan; (b)	Write SOH-CAH-TOA on the board in large letters. Say: 'These three letters are the most useful memory tool in all of trigonometry. S-O-H: Sine equals Opposite over Hypotenuse. C-A-H: Cosine equals Adjacent over Hypotenuse. T-O-A: Tangent equals Opposite over Adjacent. Copy these with the full words — not just letters.' [Give 2 minutes for copying.] Then return to the mast: 'Back to Nakuru. We know the adjacent side is 40 m and the angle is 50°. We want the opposite side — the height. Which ratio involves O and A but NOT H?' [WAIT TIME: 12 seconds. Cold-call 2 students.] 'Correct — tangent. So height = 40 × tan 50°. Open your tables to tangent. Find 50°.' [WAIT TIME: 10 seconds.] 'Auma — what does your table say?' After confirming ≈ 1.192, work through the multiplication together. Say: 'The mast is	Worked Example with Structured Release — the teacher models the complete solution to the mast problem using explicit think-aloud (naming the known, unknown, chosen ratio, table lookup, calculation), then releases learners to solve a parallel problem independently. This gradual release ensures learners can reproduce the reasoning process, not merely copy steps.	During the 3-minute independent practice, circulate and look for three specific errors: (1) choosing sin instead of tan because the student picked H instead of A as the known side — ask 'Which side is 40 m — the longest side of the triangle, or the side next to the 50° angle?'; (2) multiplying incorrectly (e.g., dividing instead of multiplying) — check equation setup; (3) reading the wrong row in the table — check by asking the student to point to the row and column they used. After the board presentation, ask the class: 'Thumbs up if your answer for the tree height matches what is on the board.' Count responses to gauge whole-class understanding.

	given sin θ = 0.643, use tables to back-calculate θ.	approximately 47.7 metres tall. We just answered the driving question — with maths alone, no ladder, no drone.' [Pause for effect.] 'Now try the Kisumu tree problem on your own — 3 minutes.' Circulate, then cold-call one student to show working on the board. For problem (b), demonstrate back-calculating: 'Find 0.643 in the sine column — what angle is it opposite?' Confirm θ ≈ 40°.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to the Pythagorean theorem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board (DQB) displayed at the front of the room. Working as a whole class, they add two new sticky-note categories: 'What we now KNOW' (green notes) and 'New questions we have' (yellow notes). Each learner writes at least one green note stating a specific thing they learned today (e.g., 'tan θ = O/A', 'tan 50° ≈ 1.192', 'the mast is about 47.7 m tall') and one yellow note with a new question (e.g., 'What if I only know the height and want the angle?', 'Why does sin + cos stay close to 1?', 'Can these ratios work for angles bigger than 90°?'). Notes are posted and the teacher reads several aloud, grouping related questions.	Point to the DQB and say: 'In Lesson 1 we asked: what tool do we need? Today we built part of that tool. Let us update our board.' Distribute green and yellow sticky notes. Say: 'Green note: write ONE specific thing you know now that you did not know at the start of today's lesson — be precise, include a formula or a number. Yellow note: write ONE new question this lesson has raised for you. You have 90 seconds.' [Allow 90 seconds.] Invite learners to post their notes. Read 3–4 green notes aloud: 'Njoroge says tan 50° ≈ 1.192 and the mast is 47.7 m. That is a direct answer to our driving question — well done.' Read 2–3 yellow notes: 'Zawadi asks — what if I know the height but not the angle? That is called finding an inverse trig ratio — we will get to that.' Circle the yellow notes that connect to upcoming lessons and say: 'These questions are exactly where our storyline goes next.'	Know/Wonder Board Update — systematically distinguishing 'established knowledge' from 'productive new questions' trains learners to see science and mathematics as a dynamic, growing enterprise rather than a fixed set of facts. Public display of yellow notes validates curiosity and previews future lessons.	Read green notes as they are posted. A note that says 'tan = O/A' with no context signals surface memorisation; a note that says 'tan 50° = 1.192 so the mast is 40 × 1.192 = 47.7 m' signals deep integration. Collect the notes at the end of class and tally: (a) How many learners correctly stated a ratio definition with formula? (b) How many connected the ratio to the mast problem? Use this data to decide whether Lesson 3 needs a 5-minute recap opener.
Model Building Phase 📹 **VIDEO:** Intro to the Pythagorean theorem	Learners open their exercise books to the annotated right-angled triangle diagram they drew in Lesson 1 (with sides labelled O, A, H). They now add a second layer to this model: alongside each side-pair, they write the ratio name	Say: 'Open to your Lesson 1 triangle model. We are going to upgrade it. In Lesson 1 it had labels — O, A, H. Today it gets the ratio formulas AND the actual mast numbers. Add sin θ = O/H, cos θ = A/H, tan θ = O/A to the correct	Cumulative Model Revision — returning to and annotating the same diagram across lessons externalises the growing understanding as a visible, evolving artefact. Learners experience knowledge as building	Circulate during the 4-minute model revision and check three things: (1) Are the three ratio formulas written next to the correct side-pairs (not randomly placed)? (2) Does the mast triangle show θ = 50° at the observer's position, 40

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and formula — 'sin θ = O/H' next to the O and H sides, 'cos θ = A/H' next to A and H, 'tan θ = O/A' next to O and A. They also add the specific mast values: θ = 50°, A = 40 m, O = ? and write the full equation 'tan 50° = O/40, therefore O = 40 × 1.192 ≈ 47.7 m' inside or beside the triangle. Learners write one sentence at the bottom of the diagram explaining in their own words what the model now represents.	side-pairs. Then sketch the mast triangle next to it with θ = 50°, A = 40 m, and show the full calculation for O. Finally, write one sentence: what does this model now allow you to do?' [Allow 4 minutes.] Circulate and check that ratio labels are placed next to the correct sides. After 4 minutes, cold-call one learner to read their sentence aloud. Say: 'This model will keep growing — next lesson we add the relationship between sines and cosines of complementary angles. By the end of the unit, this triangle will solve any height or distance problem in Kenya.'	rather than replacing, and the mast values embedded in the model maintain the direct link to the anchoring phenomenon throughout the unit.	m as the horizontal base, and the calculation tan 50° × 40? (3) Does the written sentence use the word 'ratio' or 'angle' and connect to a real-world action (measuring, calculating, finding height)? Collect three exercise books at random to review model quality as exit evidence.

D. TEACHER REFLECTION

1. When learners switched reference angles on the same triangle during the Observe Phase, did the majority correctly re-label O and A? If not, what specific intervention — a physical gesture, a colour-coding strategy, a verbal cue — might make the 'label shift' more concrete in the next lesson?\n2. Did the transition from 'computed ratio' to 'table value' feel smooth, or did learners struggle to navigate the mathematical table? Do I need to build a dedicated 5-minute 'reading the table' drill at the start of Lesson 3?\n3. When the mast height (≈ 47.7 m) was revealed for the first time, did learners show genuine engagement — surprise, satisfaction, connection? If the moment felt flat, how can I stage it more dramatically (e.g., having a learner write the final answer on the board themselves)?\n4. Looking at the green sticky notes on the DQB, what proportion of learners stated a ratio formula correctly versus a vague statement like 'I learned about triangles'? What does this tell me about the depth of definitional understanding achieved today?\n5. Did the two independent practice problems (Kisumu tree and back-calculating an angle) expose any systematic errors in ratio selection or table use that I need to address explicitly before moving to complementary angles in Lesson 3?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured sides of right-angled triangles they drew, computed three side-pair ratios for different reference angles, contributed data to a class table on the board, and then verified their computed ratios against mathematical tables — seeing that table values match measured ratios closely, confirming that the ratios depend only on the angle.
What did I learn?	The three trigonometric ratios have formal names and definitions: sin θ = opposite ÷ hypotenuse (SOH), cos θ = adjacent ÷ hypotenuse (CAH), tan θ = opposite ÷ adjacent (TOA). The labels 'opposite' and 'adjacent' change when the reference angle changes. Mathematical tables and calculators give the exact value of any ratio for any acute angle, and can be used in reverse to find an angle from a known ratio.
How does this explain the phenomenon?	We now have the specific numerical tool that was missing in the phenomenon. The student standing 40 m from the Nakuru mast at a 50° angle can use tan 50° ≈ 1.192 to calculate: height = 40 × 1.192 ≈ 47.7 m — no ladder, no drone, no guessing. The tangent ratio is the mathematical bridge between the one known side and the unknown height.

LESSON 3 (80 minutes): Sines and Cosines of Complementary Angles: sin θ = cos(90° − θ)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners discover and prove that sin θ = cos(90° − θ) and cos θ = sin(90° − θ), and apply this relationship to simplify trigonometric expressions and solve for unknown angles — deepening their toolkit for solving the mast height problem.
Knowledge	Learners know that: (a) in any right-angled triangle the two acute angles are complementary (sum to 90°); (b) sine and cosine of complementary angles are equal to each other (sin θ = cos(90° − θ)); (c) this relationship can be derived geometrically from the same right-angled triangle viewed from both acute angles.
Skills	Learners are able to: (a) generate and complete a table of complementary angle pairs and their sine/cosine values using mathematical tables; (b) identify and articulate the complementary angle pattern from data; (c) apply a geometric proof to explain why the pattern holds; (d) use the relationship sin θ = cos(90° − θ) to simplify expressions and find unknown angles.
Attitudes	Learners appreciate that mathematical patterns discovered from data can always be explained by underlying structure, and value the efficiency that the complementary angle relationship brings to solving real-world height-and-distance problems.
Key Inquiry Question	1. What is trigonometry? 2. How do we use trigonometry in real-life situations?
Purpose in Storyline	In Lessons 1–2 learners defined sine, cosine, and tangent and used mathematical tables to find the height of the Nakuru mast (tan 50° × 40 m ≈ 47.7 m). Today they notice a surprising symmetry: if the mast problem were viewed from the complementary angle (40°), sin 40° equals cos 50° exactly. This lesson gives learners a new shortcut — and deepens their understanding of why the ratios work — so that in later lessons they can move fluently between elevation and depression angles and choose the most efficient ratio.
Safety Notes	No physical hazards. Ensure learners handle mathematical tables and rulers carefully. If printed worksheets are used, distribute and collect them in an orderly manner to avoid disruption.

B. LESSON OVERVIEW
Learners open by revisiting the Nakuru mast photograph and a quick recall question about sine and cosine from Lesson 2. In the Predict phase they guess whether sin 30° and cos 60° are equal. In the Observe phase, working in pairs, they complete a structured table of six complementary angle pairs (20°/70°, 25°/65°, 30°/60°, 35°/55°, 42°/48°, 15°/75°) using mathematical tables, recording sine and cosine values to four decimal places. In the Explain phase the teacher guides them to articulate the pattern verbally, then demonstrates a geometric proof using a single right-angled triangle labelled from both acute angles — showing that the side that is 'opposite' for θ is 'adjacent' for (90° − θ), and vice versa. Learners then practise applying sin θ = cos(90° − θ) to simplify expressions and solve for unknown angles. The lesson closes with DQB updates and a cumulative model revision connecting the new relationship to the mast problem.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict	Learners are shown the Nakuru mast photograph again and reminded: 'In Lesson 2 we found	Display the mast photograph. Say: 'We found the mast height using the 50° angle at the observer's	Think-Pair-Predict: activates prior knowledge of sine and cosine definitions, creates cognitive	Circulate during the 2-minute silent prediction phase. Observable indicator: learner has written a

📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the mast is about 47.7 m tall using tan 50°. Today we look at the same triangle differently.' The teacher writes two values on the board without context: sin 30° = ? and cos 60° = ? Learners individually write a prediction in their exercise books — are these values equal, nearly equal, or completely different? They also predict what they would get if they looked up sin 40° and cos 50°. After 2 minutes, learners share predictions with a neighbour (Think-Pair).	feet. But what if we could stand at the top and look down at 40° — the complementary angle? Would our sine and cosine values be related?' Write on the board: 'PREDICTION: Is sin 30° equal to cos 60°? YES / NO / NOT SURE — write your reason.' Allow 2 minutes of silent individual thinking (WAIT TIME: do not take answers yet). Cold-call 3 learners for their predictions and reasons — choose one YES, one NO, one NOT SURE. Record all three on the board under 'Our Predictions'. Say: 'By the end of today, you will be able to prove which of you is right — and explain WHY using nothing but a triangle.'	tension (learners are genuinely uncertain whether a pattern exists), and gives every learner a personal stake in the data-gathering activity that follows.	specific numerical estimate (e.g., 'sin 30° ≈ 0.5, cos 60° ≈ 0.5, so YES') or a reasoned guess referencing the triangle definition. Learners who write 'I don't know' with no attempt need immediate paired support in Phase 2.
Phase 2 — Observe 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Working in pairs, learners receive or copy a structured two-column table with six complementary angle pairs. Using their mathematical tables, they look up and record sin θ, cos θ, sin(90° − θ), and cos(90° − θ) for each pair to four decimal places. Pairs are then asked to write one sentence describing any pattern they notice. A class data-share follows where each pair reads out one row of values and the teacher populates a master table on the board. The class jointly inspects the completed master table. Angle pairs used: – θ = 20°, (90°−θ) = 70° – θ = 25°, (90°−θ) = 65° – θ = 30°, (90°−θ) = 60° – θ = 35°, (90°−θ) = 55° – θ = 42°, (90°−θ) = 48° – θ = 15°, (90°−θ) = 75° Learners also check their Lesson 2	Distribute or dictate the table structure. Say: 'Use your mathematical tables. Work in pairs. Fill in sin θ and cos(90° − θ) for each row. You have 12 minutes. I will be circulating — if you finish early, check whether sin(90° − θ) equals cos θ as well.' Set a visible timer. Circulate — check that learners are reading the sine column and not the tangent column (a common error). After 12 minutes, say: 'Pens down. Each pair, tell me one row.' Cold-call all 6 pairs in sequence to build the master table on the board. After the table is complete, allow 30 seconds of silent inspection (WAIT TIME). Then ask: 'In one sentence — what pattern do you see in this table?' Cold-call 4 learners. Accept and paraphrase: 'So what you are all telling me is: sin θ = cos(90° − θ). Let us write that formally.' Write on the board: sin θ = cos(90° − θ) and cos θ = sin(90° − θ).	Data-Table Pattern Recognition (Inductive Reasoning): learners generate real numerical evidence before being told the rule. Seeing the equality hold across six different angle pairs — including their own mast-problem pair (sin 40° = cos 50°) — makes the generalisation feel discovered rather than imposed. Recording the mast pair explicitly connects the new pattern to the anchoring phenomenon.	Inspect pairs' tables during circulation. Observable indicators: (1) four decimal places recorded correctly for at least 4 of 6 rows; (2) sin 40° = cos 50° = 0.6428 identified correctly; (3) the one-sentence pattern description uses the words 'equal', 'same', or 'complement'. Pairs with systematic errors (e.g., all cosine values wrong) are using the wrong column in the table — redirect immediately by pointing to the column header.

	result: sin 40° vs cos 50° — the mast's complementary pair.			
Phase 3 — Explain 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher draws a single right-angled triangle on the board, labelling the right angle C, angle A = θ, angle B = (90° − θ), hypotenuse = c, side opposite A = a, side adjacent A = b. Learners first write sin θ and cos θ in terms of a, b, c. The teacher then says: 'Now stand at corner B instead of corner A — what is sin(90° − θ)?' Learners rewrite the ratios from B's perspective and discover that sin(90° − θ) = b/c = cos θ. Learners then practise three categories of application: (A) Simplify: express cos 70° as a sine; express sin 55° as a cosine. (B) Solve for θ: sin(2θ + 10°) = cos(θ + 20°) → find θ. (C) Mast link: 'A surveyor in Kisumu stands 60 m from a building. The angle of elevation is 38°. Write the height using tan 38°, then rewrite cos 38° in terms of a sine — which angle?' Learners work individually for 8 minutes, then compare with a neighbour.	Draw the triangle slowly, naming each side aloud. Say: 'I am going to label this triangle twice — once from A's point of view, once from B's. Your job is to write the sine and cosine ratios each time.' After learners write individually (WAIT TIME: 15 seconds), cold-call 2 learners for sin θ and cos θ from A. Then say: 'Now, from B — what is the opposite side? What is the adjacent side?' Cold-call 2 different learners. Write the ratios side by side so learners can see sin(90° − θ) = b/c = cos θ. Say: 'This is not a coincidence — it is true for every right-angled triangle, every time. This is our proof.' Transition to practice: 'Now apply this. Work the three sets individually. You have 8 minutes.' After 8 minutes: cold-call for Part B answer (θ = 20°) — ask the learner to show each algebraic step aloud. For Part C, explicitly say: 'How does this connect back to our mast in Nakuru? Could a surveyor on the ground use either the 50° or the 40° angle to describe the same mast height situation?'	Geometric Proof as Sensemaking (Perspective Shift): by physically re-labelling the same triangle from the complementary angle's viewpoint, learners see that the pattern is not coincidence but a structural consequence of triangle geometry. This moves understanding from 'the table says so' to 'I can see WHY'. The Kisumu building extension maintains the phenomenon thread.	Collect Part B answers: observable indicator is θ = 20° with correct algebraic steps shown (2θ + 10° + θ + 20° = 90° → 3θ = 60° → θ = 20°). Learners who arrive at a different answer have likely set the two angles equal rather than summing to 90° — address this as a whole-class misconception correction before moving to the DQB phase. For Part A: cos 70° = sin 20° and sin 55° = cos 35° are the expected answers.
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Each learner writes on a sticky note (or in their book if notes are unavailable) one statement and one remaining question. The statement begins: 'I now understand that…' and the remaining question begins: 'I am still wondering…' Learners post or read out their notes. The teacher organises the responses into two columns on the DQB: WHAT WE NOW KNOW and WHAT WE STILL NEED. The class reviews the driving question — 'How can	Say: 'Look at our driving question on the board. In Lesson 1 we had no tool at all. In Lesson 2 we got sine, cosine, and tangent. What has today added?' Allow 20 seconds of silent thinking (WAIT TIME). Cold-call 3 learners. Guide toward: 'We now know that sin 50° = cos 40° — so if a surveyor can measure the angle of elevation OR the angle of depression, they get the same answer.' Add to the DQB: 'sin θ = cos(90° − θ): one triangle, two angles, same ratios	Driving Question Board — Cumulative Knowledge Tracking: the DQB makes the growing understanding visible across the unit. Learners see that each lesson closes a gap and opens new questions, modelling authentic scientific and mathematical inquiry. Connecting today's result to the mast phenomenon (sin 50° = cos 40°) anchors the abstraction.	Read all sticky notes / written statements. Observable indicators of understanding: learner correctly states the complementary relationship in words or symbols; learner can give a concrete example (e.g., 'sin 65° = cos 25°'). Observable indicator of a remaining gap: learner confuses the relationship direction (writing 'sin θ = sin(90° − θ)') — note these learners for targeted follow-up at the start of Lesson 4.

[Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	we use angles and one side of a right-angled triangle to find any unknown height or distance?' — and discusses how today's lesson adds a new tool: the complementary angle relationship means we can always convert between sine and cosine, giving surveyors and engineers flexibility in choosing which angle to work from.	swapped.' Ask: 'What questions does this raise? Could there be special angles where sin θ = cos θ exactly?' Accept all questions without answering — write the most mathematically productive ones on the board. (This seeds Lesson 4 on special angles: 45° is exactly where sin = cos.)		
Phase 5 — Model Building 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative right-angled triangle diagram started in Lesson 1 (the Nakuru mast sketch with angle of elevation 50°, base 40 m, height h). They now annotate it with two new features: (1) they label the complementary angle of depression (40°) at the top of the mast — the angle a hawk perched on the mast would see down to the observer; (2) they write next to the diagram: sin 50° = cos 40° = h/hypotenuse and cos 50° = sin 40° = 40/hypotenuse, showing both angles produce the same ratios in swapped roles. Below the diagram, learners write one sentence explaining why a Kenya Power engineer could use either the 50° elevation angle at the base or the 40° depression angle at the top to calculate the same mast height.	Say: 'Take out your mast diagram from Lesson 1. We are going to add to it — our model is growing as our knowledge grows.' Project or draw the reference mast diagram. Say: 'Add the 40° angle at the top of the mast — the angle a bird would look down at the observer. Then write the complementary relationship for both angles next to the diagram.' Allow 5 minutes. Circulate and check that learners are placing the 40° angle correctly (at vertex between the mast height and the hypotenuse line of sight). Cold-call 2 learners to read their explanatory sentence aloud. Close with: 'Next lesson we look at very special angles — 30°, 45°, 60° — where the ratios have exact, beautiful values. Today's relationship will help us get there faster.'	Cumulative Annotated Model Revision: each lesson, learners add one new layer of mathematical understanding to the same physical diagram. This makes conceptual growth concrete and visible. Annotating the mast diagram with the complementary angle reinforces that sin θ = cos(90° − θ) is not just an algebraic trick — it describes real geometry in a real Kenyan context.	Inspect diagrams before learners leave or at the start of Lesson 4. Observable indicators: (1) 40° correctly placed at the top of the mast; (2) sin 50° = cos 40° written correctly next to the diagram; (3) the explanatory sentence correctly references 'complementary angles' or 'the angle at the top and the angle at the base add to 90°'. Diagrams missing the top angle or reversing the equality are flagged for individual feedback.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners genuinely discover the pattern from the table, or did they wait for me to tell them? If learners were passive, next time ask pairs to write their pattern sentence BEFORE the whole-class share. (2) Did the geometric proof land — could learners re-explain WHY sin θ = cos(90° − θ) without looking at notes? If not, add a second triangle labelling exercise. (3) Were learners able to set up and solve sin(2θ + 10°) = cos(θ + 20°) algebraically, or did they guess? Algebraic fluency here is essential before Lesson 5 (elevation and depression). (4) Did the DQB surface the question 'Is there an angle where sin θ = cos θ?' — if yes, Lesson 4 on special angles has a perfect entry point. If not, plant that question explicitly at the start of Lesson 4. (5) Check: did all learners correctly identify sin 40° = cos 50° = 0.6428 in the mast context? This is the bridge to the anchoring phenomenon and must be secure.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners completed a table of six complementary angle pairs (e.g., 30° & 60°, 42° & 48°) using mathematical tables, recording sine and cosine values to four decimal places. They observed that $\sin \theta = \cos(90° - \theta)$ and $\cos \theta = \sin(90° - \theta)$ held true across all pairs, including the mast-problem pair $\sin 40° = \cos 50° = 0.6428$.
What did I learn?	Learners learned that in any right-angled triangle, the sine of one acute angle equals the cosine of the other, because the side that is 'opposite' for one angle is 'adjacent' for the complementary angle. This was proved geometrically by re-labelling the same triangle from both vertices. The relationship $\sin \theta = \cos(90° - \theta)$ is a structural property of right-angled triangles, not a numerical coincidence.
How does this explain the phenomenon?	Learners applied the complementary angle relationship to: (a) rewrite trigonometric expressions (e.g., $\cos 70° = \sin 20°$); (b) solve for unknown angles algebraically (e.g., $\sin(2\theta + 10°) = \cos(\theta + 20°) \rightarrow \theta = 20°$); and (c) annotate the Nakuru mast diagram with both the 50° elevation angle and the complementary 40° depression angle, writing the swapped sine-cosine ratios for each — showing that a Kenya Power engineer can work from either angle to find the same mast height.

LESSON 4 (80 minutes): The Hidden Link: Discovering tan θ = sin θ / cos θ	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners discover and formalise the identity tan θ = sin θ / cos θ by computing ratios from their own data tables, then apply the identity to find missing trigonometric values — moving one step closer to solving the Nakuru mast problem using only one side and one angle.
Knowledge	Learners will know that tan θ = sin θ / cos θ for any acute angle θ, and that knowing any two of the three primary trigonometric ratios allows them to determine the third.
Skills	Learners will be able to: (i) compute sin θ / cos θ for a set of angles and compare results with recorded tan θ values; (ii) apply the identity tan θ = sin θ / cos θ to find a missing ratio when two are known; (iii) verify calculator outputs using the identity; (iv) solve short problems such as 'given sin θ = 3/5, find cos θ and tan θ without a calculator.'
Attitudes	Learners appreciate that mathematical relationships are discovered through systematic observation of patterns, not merely stated as rules — and that this same discipline of noticing patterns is what allows engineers and surveyors in Kenya to trust their instruments.
Key Inquiry Question	How do we use trigonometry in real-life situations?
Purpose in Storyline	In Lesson 2 learners built a table of sin θ, cos θ, and tan θ values. In Lesson 3 they explored complementary-angle relationships. Now in Lesson 4 they mine that same table for a new pattern — the ratio sin θ / cos θ equals tan θ — and formalise it as an algebraic identity. This identity is a critical internal tool: it means that in the mast problem, if a surveyor in Nakuru measures sin 50° and cos 50° from tables, they can instantly recover tan 50° to compute height, even if their calculator's tan button fails. The lesson also sets up Lesson 5's work on special angles, where the identity will be used to verify exact values.
Safety Notes	No physical hazards. Remind learners to handle mathematical tables carefully and to double-check calculator mode (DEG not RAD) before verifying outputs. Emphasise that sharing one calculator per pair is acceptable — patience and turn-taking are values in action.

B. LESSON OVERVIEW

Learners open by re-examining the Nakuru mast photograph and confronting a new puzzle: if a surveyor's calculator tan button breaks, can they still find tan 50° using the other buttons? In the Predict phase learners guess whether sin θ / cos θ might equal tan θ. In the Observe phase learner pairs divide sin θ by cos θ for six angles drawn from their Lesson 2 tables and compare the quotients with their recorded tan θ values, noticing exact agreement. In the Explain phase the teacher formalises the algebraic proof of tan θ = sin θ / cos θ from the definitions (opposite/hypotenuse ÷ adjacent/hypotenuse), then learners practise using the identity in both directions. The DQB is updated with the new identity. In the Model Building phase learners annotate their cumulative right-triangle diagram to show sin θ / cos θ as a third labelled ratio pathway, and solve the short problem: given sin θ = 3/5, find cos θ and tan θ without a calculator, connecting back to the mast scenario.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (≈ 12	Learners look at the projected Nakuru mast photograph again. The teacher poses a new version	Project the mast photograph. Say verbatim: 'We now know sin, cos, and tan separately — but are they	Prediction Journaling + Claim–Evidence–Reasoning (CER) setup. By committing a written prediction	Teacher circulates during individual writing (2 minutes) and reads over shoulders. Observable

minutes) 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	of the driving question: 'Imagine the Kenya Power engineer on site has sin 50° = 0.766 and cos 50° = 0.643 in her field notebook, but her calculator's tan button is broken. Can she still find tan 50° — and therefore the mast height?' Learners write a 2-sentence individual prediction in their exercise books: (1) Will sin 50° divided by cos 50° give the same answer as tan 50°? (2) How confident are you, and why? Learners then share predictions in pairs before a whole-class show of hands: 'Hands up if you predicted YES.' Teacher records the class split on the board without evaluating.	strangers to each other, or are they secretly connected?' Pause 12 seconds (WAIT TIME). Cold-call 3 learners for their written predictions before the pair-share. Record on the board: 'Class prediction: [X] say YES, [Y] say NO, [Z] say MAYBE.' Say: 'We are going to let the numbers from our own tables tell us the truth. Scientists and engineers in Kenya do not guess — they test.'	before seeing the data, learners activate prior knowledge of the ratio definitions and create a personal stake in the outcome. The whole-class tally makes disagreement visible and motivates the investigation.	indicator: learner's written sentence references the definitions of sin and cos (opposite/hypotenuse, adjacent/hypotenuse) — not just a gut feeling. Learners who write only 'I think yes' without reasoning are prompted: 'Write down what sin 50° actually means as a fraction of sides — that will help you decide.'
Phase 2 — Observe (≈ 20 minutes) 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs. Each pair retrieves their Lesson 2 data table (or uses a printed copy provided by the teacher) listing sin θ, cos θ, and tan θ for θ = 20°, 30°, 40°, 50°, 60°, 70°. Each pair adds a fourth column labelled 'sin θ ÷ cos θ' and uses a calculator to compute and record the quotient to 3 decimal places for each of the six angles. They then compare the new column with their tan θ column entry by entry and record an observation sentence: 'sin θ ÷ cos θ is [always / sometimes / never] equal to tan θ.' After 12 minutes of pair work, pairs join into groups of four to compare tables and resolve any discrepancies caused by rounding. Each group nominates a reporter to share one finding with the class.	Distribute printed data-table sheets to any learners whose Lesson 2 tables are incomplete. Circulate actively; after 5 minutes cold-call 2 pairs: 'Read me your value for sin 30° ÷ cos 30° and your recorded tan 30° — what do you notice?' (expected: 0.577 and 0.577). After groups have compared tables, ask whole class: 'Raise your hand if every single row in your table matched.' Pause 10 seconds (WAIT TIME). Say: 'This is not a coincidence. Mathematicians call a relationship that is always true an identity. Today we will prove why it must be true.' Write on the board: 'tan θ = sin θ / cos θ — to be proved.'	Data-Pattern Recognition (tabular investigation). By generating the ratio sin θ / cos θ independently and comparing it to a separately measured tan θ, learners experience the identity as an empirical discovery before it is formalised. This mirrors how Kenyan survey engineers cross-check field measurements against reference tables — building trust in mathematical tools through verification.	Observable indicator: completed fourth column with values matching tan θ column to 2 decimal places for all six angles, plus a written observation sentence. Teacher flags any pair whose values diverge (likely a calculator-mode error — check DEG/RAD) and uses the error as a teachable moment: 'This is exactly why engineers always verify their calculator mode before computing angles on a construction site in Nairobi or Kisumu.'
Phase 3 — Explain (≈ 22 minutes) 📹 **VIDEO:** [Trigonometry]	The teacher leads a structured proof at the board. Learners copy the proof into their notes and annotate each step with a marginal note explaining in their own words	Write a labelled right-angled triangle on the board with sides: opposite = O, adjacent = A, hypotenuse = H. Write: 'sin θ = O/H, cos θ = A/H.' Say: 'Now I	Algebraic Proof + Graduated Practice (I Do — You Do — We Check). The teacher models the proof explicitly, then learners immediately apply it in three	Observable indicator for Problem A: learner correctly writes cos θ = 4/5 (using the 3-4-5 Pythagorean triple) and tan θ = 3/4. Common error to watch: learner divides 3 by

[Word Problems Principles - Basic](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometry Word Problems (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	why the step is valid. After the proof, learners attempt three graduated practice problems individually: (A) Given sin θ = 3/5, and knowing that $\sin^2\theta + \cos^2\theta = 1$ (previewed, not yet formally derived), find cos θ, then use the identity to find tan θ. (B) Given tan 35° = 0.700 and cos 35° = 0.819, use the identity to find sin 35°, then verify with a calculator. (C) A Kenya Power engineer records that the angle of elevation to the top of the Nakuru mast is 50°. She has sin 50° = 0.766 and cos 50° = 0.643. Use tan θ = sin θ / cos θ to find tan 50°, then compute the height of the mast given horizontal distance = 40 m. Learners check each other's work in pairs before whole-class debrief.	divide sin θ by cos θ. Watch every step.' Write: '(O/H) ÷ (A/H) = (O/H) × (H/A) = O/A = tan θ.' Circle the result. Say: 'The hypotenuse cancelled — it disappeared — leaving us O/A, which is exactly the definition of tan θ. This is not magic; it is algebra.' Pause 12 seconds (WAIT TIME). Cold-call 4 learners: 'Tell me in your own words why H cancelled.' For Problem A, do not solve it — say: 'Spend 4 minutes individually, then compare with your partner.' After pair-check, cold-call one pair to present Problem C at the board. Emphasise: 'We have now answered the broken-calculator question for the Kenya Power engineer. She can always recover tan θ from her notebook.'	problems of increasing proximity to the anchoring phenomenon. Problem C closes the loop to the Nakuru mast, making the identity feel purposeful rather than abstract.	5 twice instead of finding the missing side — redirect with: 'Draw the triangle. Label the side you know. Which side is missing?' For Problem C: learner should reach height ≈ 47.7 m. Any answer outside 47–48 m warrants a calculator-mode check.
Phase 4 — Driving Question Board (DQB) Update (≈ 10 minutes) 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometry Word Problems (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the ORANGE note they write one thing they can now do that they could not do at the start of the lesson (e.g., 'I can find tan θ from sin θ and cos θ without a separate table'). On the BLUE note they write one new question the lesson raised (e.g., 'Does this identity work for 90°? Cos 90° = 0, so would we be dividing by zero?'). Learners post both notes on the class Driving Question Board in the appropriate columns: 'We Now Know' and 'We Still Wonder'. The teacher reads three blue notes aloud and annotates them: questions about 90° and about going backwards (given tan, find sin and cos) are flagged as 'coming in Lesson 5 and Lesson 6.'	Point to the DQB and say: 'At the start of this unit, we asked how the Nakuru engineer could find the mast height using only one side and one angle. Each lesson, we add a new tool to her toolkit. Today's tool is the identity. Let us record that.' After learners post notes, read 3 blue notes aloud without identifying authors. For the 'cos 90° = 0' question, say: 'Excellent — that is a mathematician's instinct. We call that a undefined ratio, and we will meet it in Lesson 5.' Do NOT resolve it now — let curiosity persist on the board.	Driving Question Board — Know/Wonder Sort. Making 'what I now know' and 'what I still wonder' public builds metacognitive awareness and maintains the narrative thread of the unit. Flagging blue-note questions for future lessons models how mathematical inquiry is cumulative — exactly as it is in real survey projects across Kenya's infrastructure sector.	Observable indicator: orange note contains a specific capability statement referencing the identity (not just 'I learned tan'). Blue note contains a genuine mathematical question (not a complaint or unrelated thought). Teacher mentally notes learners who cannot articulate an orange-note capability — those learners need a brief 3-minute recap at the start of Lesson 5.
Phase 5 — Model Building (≈ 16	Learners retrieve their cumulative right-triangle model diagram begun in Lesson 1 and updated in	Say: 'Your model diagram is growing. Every lesson we add a new layer — just like a building in	Cumulative Model Revision. Adding the identity as a visual pathway on the right-triangle	Observable indicator: curved arrow is correctly placed and labelled; the 3-4-5 example box is

minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Lessons 2 and 3. Today they add two new annotations: (1) A curved arrow from the 'sin θ' label to the 'cos θ' label labelled '÷ gives tan θ' — showing the identity as a pathway on the diagram. (2) A worked example box beside the diagram showing the 3-4-5 triangle (sin θ = 3/5, cos θ = 4/5, tan θ = 3/4) with each ratio confirmed by the identity. At the bottom of the model page, learners write a one-sentence answer to the driving question as it stands today: 'Using tan θ = sin θ / cos θ and the horizontal distance of 40 m, the height of the Nakuru mast is approximately ___ m.' Learners who finish early attempt the extension: 'If the mast were twice as tall but the engineer stood at the same distance, what angle would she measure? Use tan θ = height / distance and work backwards.'	Nairobi grows floor by floor. Today we are adding the identity as a bridge between our three ratios.' Circulate and check that the curved arrow is correctly labelled. After 8 minutes, cold-call 3 learners to read their driving-question sentence aloud, checking that the height value is approximately 47.7 m. For the extension, prompt: 'What trigonometric ratio links the height and the horizontal distance directly?' (tan). 'So if height doubles to 95.4 m and distance stays 40 m, what is tan θ?' (≈ 2.385). 'Can you find that angle from your table or calculator?' Close the lesson: 'Next lesson we will meet special angles — 30°, 45°, 60° — where sin, cos, and tan have exact, beautiful values that engineers memorise. Today's identity will help us verify them.'	diagram makes the relationship structural — learners can literally see sin and cos feeding into tan. The driving-question sentence forces every learner to produce a numerical answer, connecting the abstract identity to the concrete mast-height computation and closing the predict–observe–explain cycle.	completed with all three ratios confirmed. Driving-question sentence includes a numerical height (approximately 47.7 m). Extension indicator: learner correctly sets up tan θ = 95.4 / 40 and uses inverse tan or table to find θ ≈ 67.2°, demonstrating readiness for angles-of-elevation work in Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Observe phase, did all pairs successfully complete the fourth column of their data table? If not, was the root cause a calculator-mode error (DEG/RAD), an incomplete Lesson 2 table, or a conceptual gap in dividing fractions? Plan a 3-minute warm-up for Lesson 5 if more than 25% of learners had incomplete tables. (2) In the Explain phase, did learners articulate why H cancelled in the algebraic proof in their own words, or did they copy the steps without understanding? If cold-called learners could not explain the cancellation, consider starting Lesson 5 with a brief 'spot-check quiz': write sin θ / cos θ on the board and ask learners to simplify it from definitions. (3) Did Problem C (the mast height) produce the expected answer of ≈ 47.7 m from the majority of learners? If not, note whether the error was in the identity application or in the final multiplication (tan 50° × 40 m) — these require different remediation approaches. (4) Review the blue sticky notes from the DQB. Tally how many learners raised the 'dividing by zero at 90°' question — a high count signals strong mathematical curiosity and the class may benefit from a brief discussion at the opening of Lesson 5. (5) Were Kenyan contexts (Kenya Power engineer, Nakuru mast, 3-4-5 triangle framing) sufficient to keep learners engaged, or did the algebraic proof phase feel disconnected from the phenomenon? If so, consider adding a second real-life hook in Lesson 5 — for example, a photograph of the Rift Valley transmission lines where surveyors use exactly these ratios.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners computed sin θ / cos θ for six angles (20°–70°) from their Lesson 2 data tables and found that the quotient matched their recorded tan θ values exactly in every row, regardless of the angle chosen.

What did I learn?	$\tan\theta = \sin\theta / \cos\theta$ for all acute angles θ. This identity follows algebraically from the definitions: (O/H) ÷ (A/H) = O/A = tan θ. Knowing any two of the three primary ratios allows us to determine the third.
How does this explain the phenomenon?	Because sin θ and cos θ are both defined using the hypotenuse as denominator, dividing them cancels the hypotenuse and leaves the ratio of opposite to adjacent — which is precisely tan θ. This is why the Kenya Power engineer can recover tan 50° from her notebook even without the tan button, and compute the Nakuru mast height as $\tan 50° \times 40$ m ≈ 47.7 m.

LESSON 5 (80 minutes): Trigonometric Ratios of Special Angles (45°, 30°, 60°, 0°, and 90°)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive and apply exact trigonometric ratios for special angles (0°, 30°, 45°, 60°, and 90°) using geometric constructions, so they can solve height-and-distance problems — including the Nakuru mast problem — without a calculator or tables.
Knowledge	Learners will know that: (a) an isosceles right-angled triangle with legs of 1 unit yields sin 45° = cos 45° = $1/\sqrt{2}$ and tan 45° = 1; (b) bisecting an equilateral triangle of side 2 units yields a 30-60-90 triangle with sides 1, $\sqrt{3}$, and 2, giving sin 30° = cos 60° = 1/2, sin 60° = cos 30° = $\sqrt{3}/2$, tan 30° = $1/\sqrt{3}$, and tan 60° = $\sqrt{3}$; (c) sin 0° = 0, cos 0° = 1, tan 0° = 0, sin 90° = 1, cos 90° = 0, and tan 90° is undefined, justified by limit arguments from their trig-tables work in Lesson 3.
Skills	Learners will be able to: (a) construct the two special triangles accurately using ruler and compasses; (b) apply Pythagoras' theorem to find exact side lengths; (c) read off and record the six special-angle ratios in exact (surd) form; (d) use the special-angle reference card to solve right-angled triangle problems — including finding the height of the Nakuru mast — without a calculator.
Attitudes	Learners will appreciate that mathematical precision (exact surd values) is more powerful and elegant than decimal approximations, and that geometry and algebra together create tools that solve real Kenyan engineering and construction challenges.
Key Inquiry Question	1. What is trigonometry? 2. How do we use trigonometry in real-life situations?
Purpose in Storyline	Lessons 1–4 gave learners the general definitions of sin, cos, and tan, plus skills in reading tables and calculators. Lesson 5 unlocks EXACT values for the five special angles. This means learners can now solve the Nakuru mast problem (50° ≈ 45° first approximation, then exactly at 60°) and any 30-60-90 construction problem — such as the slope of a ramp or the pitch of a roof — purely by reasoning, without any table or device. The special-angle card they build today becomes a permanent problem-solving tool for Lessons 6–8.
Safety Notes	Compasses have sharp points — teacher reminds learners to keep compass points directed downward onto paper/boards and never to wave or pass them open. Scissors (if used to cut the reference card) should be handled with blades closed when not in use. Ensure adequate desk space so compasses do not accidentally contact neighbours.

B. LESSON OVERVIEW

Learners work in groups of four. In the PREDICT phase they guess the exact value of sin 45° and tan 60° before any construction. In the OBSERVE phase, each group constructs (i) an isosceles right-angled triangle with legs of 1 unit and (ii) an equilateral triangle of side 2 units that they bisect into a 30-60-90 triangle; they measure and record all sides. In the EXPLAIN phase, groups apply the sine, cosine, and tangent definitions to derive exact surd values for all five special angles, verify the limit arguments for 0° and 90°, and use their values to re-solve the Nakuru mast problem. In the DQB phase, groups post refined answers and new questions to the class Driving Question Board. In the MODEL-BUILDING phase, each learner adds a completed special-angles reference card — designed in Kenyan flag colours (black, red, green, white) — to their cumulative unit model booklet. All five phases are explicitly tied back to the anchoring phenomenon: "How tall is that mast near Nakuru — and can we now find out without climbing it?"

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Phase 1 — PREDICT (10 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners sit in their groups of four. Each learner individually writes answers to three prompt questions in their exercise books BEFORE any instruction or construction: (1) 'Without using tables or a calculator, what do you think sin 45° equals? Write a fraction or decimal guess.' (2) 'A Kenya Power technician needs to install a diagonal wire brace on a 60° mast support. If the horizontal distance is 10 m, she needs tan 60°. What is your best guess for tan 60°?' (3) 'Return to the Nakuru mast: if the angle of elevation were exactly 45° instead of 50°, and the horizontal distance is still 40 m, estimate the mast height.' Learners record their reasoning, not just answers. After 3 minutes of individual thinking, groups share predictions verbally for 2 minutes — no consensus required.	Teacher displays the anchoring photograph of the Nakuru electricity transmission tower and re-reads the driving question aloud: 'Using only that horizontal distance and that angle, can we calculate how tall the mast is without ladders, drones, or guessing?' Teacher then says: 'Today we are going to unlock EXACT values — no tables needed. But first, predict.' Teacher circulates silently for 3 minutes of individual writing, offering no hints. Teacher then cold-calls EXACTLY 3 learners — one who appears confident, one hesitant, one in between — using first names: 'Akinyi, what did you write for sin 45°?', 'Mwangi, what is your guess for tan 60°?', 'Fatuma, what height did you estimate for the mast at 45°?' WAIT TIME: teacher counts silently to 12 after each question before accepting or redirecting. Teacher records all three predictions visibly on the board under the heading 'OUR PREDICTIONS — we will check these at the end.' Teacher does NOT correct predictions at this stage.	Strategy: PREDICT-FIRST ANCHORING. By committing predictions to paper before instruction, learners activate prior knowledge (Pythagoras, decimal approximations from tables in Lesson 3) and create a personal knowledge gap. When their exact derived values later confirm or correct their guesses, the cognitive dissonance deepens retention. Tying predictions explicitly to the mast problem sustains the phenomenon thread.	Teacher scans written predictions during circulation. Observable indicators of readiness: (a) learner can write any fraction or surd attempt for sin 45° — even '1/2' or '0.7' shows engagement with the ratio idea; (b) learner attempts to use tan × distance = height for the mast estimate, showing carry-over from Lesson 4. Red flag: learner leaves all three blank — teacher quietly asks 'What does sin mean? Draw me the triangle' to re-anchor before the Observe phase.
Phase 2 — OBSERVE: Geometric Construction and Measurement (22 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos	Each group receives: 1 sheet of plain A4 paper, a ruler, a compass, a protractor, and a pencil. The task is split into two constructions completed simultaneously by pairs within each group. CONSTRUCTION A (pairs 1–2 in each group): Draw an isosceles right-angled triangle. Using ruler and compass, draw a horizontal line segment AB of exactly 6 cm. At point A, construct a perpendicular of exactly 6 cm to get point C. Connect B to C to form the hypotenuse. Label all	Teacher begins by saying: 'You are going to do what Kenyan surveyors and engineers do — build the triangle on paper before building it in the field.' Teacher demonstrates ONLY the first compass step for Construction A on the board (drawing the perpendicular), then steps back: 'The rest is yours.' Teacher circulates, checking constructions every 2–3 minutes. At the 8-minute mark teacher pauses the class for 30 seconds: 'Pairs doing Construction B — has everyone	Strategy: CONSTRUCT-BEFORE-CALCULATE. Hands-on geometric construction grounds the abstract surd values in physical measurement. When learners measure BC ≈ 8.49 cm and later derive BC = 6√2 exactly, the match between measurement and algebra is a memorable 'aha' moment. Using 6 cm and 8 cm as practical lengths (rather than unit lengths) makes the construction accurate and measurable, while the ratio approach still yields the same exact trig values. The group-	Teacher checks that: (a) all four angles in Construction A sum to 180° and the right angle is correctly marked; (b) PM in Construction B is longer than QM but shorter than QP — if PM > QP, the student has made an error in bisection. Specific indicator: group mini-whiteboards show a correctly labelled table with all three sides for each triangle. If any group's BC measurement is far from 8.49 cm (e.g., 7 cm or 10 cm), teacher pauses that group and re-checks the right-angle construction before

📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	angles (45°, 45°, 90°) and all sides. Measure BC with a ruler and record it. Pairs then write: opposite, adjacent, and hypotenuse for the 45° angle at B. CONSTRUCTION B (pairs 3–4 in each group): Draw an equilateral triangle with side 8 cm (representing '2 units'). Label vertices P, Q, R. Draw the perpendicular bisector from P to the midpoint M of QR. Label the two resulting right-angled triangles. Measure PM. Label all angles (30°, 60°, 90°) and all sides (QM = 4 cm, QP = 8 cm, PM = measure and record). Each pair records their measurements in a table: Angle | Opposite | Adjacent | Hypotenuse. After 15 minutes of construction, pairs share results across their group (2 minutes) and agree on a common set of measurements. Group recorder writes the agreed values on a mini-whiteboard or large paper for display.	found PM? Measure carefully — this is the side your mast height will depend on.' Teacher uses two specific probing questions during circulation, asking each group at least one: (i) 'For your 45° triangle, what is the ratio of BC to AB? Write it as a fraction.' WAIT TIME: 12 seconds. (ii) 'In your 30-60-90 triangle, which side is the hypotenuse and which angle is opposite the shortest side?' WAIT TIME: 12 seconds. At the 20-minute mark, teacher cold-calls one group to read their measured value of BC (expected ≈ 8.5 cm for the 6-6-x triangle, i.e. 6√2 ≈ 8.49 cm) and one group to read PM (expected ≈ 6.93 cm for the 4-x-8 triangle, i.e. 4√3 ≈ 6.93 cm). Teacher writes both on the board and says: 'Hold onto these numbers — in Phase 3 we will explain exactly WHY they are what they are.'	display step makes individual measurements social and self-correcting.	Explain phase.
Phase 3 — EXPLAIN: Deriving Exact Values and Solving the Mast Problem (22 minutes) 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES	PART A — Exact derivation (12 minutes): Each learner individually applies the definitions sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, tan θ = opposite/adjacent to their two constructed triangles. For the 45° triangle (legs = 1 unit, hypotenuse = √2): learners write sin 45° = 1/√2, cos 45° = 1/√2, tan 45° = 1. For the 30-60-90 triangle (short leg = 1, long leg = √3, hypotenuse = 2): learners write all six values. For 0° and 90°, learners look back at the trig-table columns from Lesson 3 and complete: 'As the angle approaches 0°, the opposite side approaches ___, so sin 0° = ___. As the angle approaches 90°, the opposite side approaches the	Teacher opens Part A by writing on the board: 'Our rule from Lesson 1: sin θ = O/H, cos θ = A/H, tan θ = O/A. Your triangles are already drawn. Now just read off the ratios.' Teacher sets a 12-minute individual silent work timer. During this time teacher circulates and targets two common errors: (i) learners writing sin 45° = 6/8.49 instead of 1/√2 — teacher prompts: 'You used 6 cm for the leg. What unit-length ratio does that give? Simplify.' WAIT TIME: 12 seconds. (ii) learners confusing which leg is opposite 30° vs 60° — teacher prompts: 'Point to the angle that is 30°. Now point to the side directly across from it — that is your opposite.' WAIT TIME: 12	Strategy: WORKED-EXAMPLE FADE. Learners first derive values independently (high cognitive demand), then immediately apply them to the phenomenon-anchoring mast problem (transfer). The explicit comparison of the 50° decimal answer (Lesson 1) with the 60° surd answer (today) creates a cross-lesson sensemaking thread: learners see that special angles give elegant exact answers, while non-special angles need tables or calculators. This contrast reinforces WHY both tools (exact values and tables) are needed.	Observable indicators: (a) learner's 6-column table has all correct exact values (no decimal approximations in the sin/cos/tan columns for special angles); (b) learner correctly writes tan 90° as 'undefined' with a brief reason ('the adjacent side has length 0, and division by zero is undefined'); (c) mast working shows height = 40 × tan 60° = 40√3 and a correct decimal equivalent. Red flag for misconception: learner writes tan 45° = √2 — teacher re-asks 'What is O/A for the isosceles triangle where both legs equal 1?'

for similar readings	hypotenuse, so sin 90° = ___.' Learners record all values in a 6-column table: Angle | sin | cos | tan. PART B — Mast application (6 minutes): Teacher poses the adapted mast problem on the board: 'Suppose the surveyor in Nakuru measures the angle of elevation to the top of the Kenya Power mast as exactly 60°, from a point 40 m away. Using only your special-angles card, calculate the height of the mast.' Learners work individually then compare with a partner. Expected: height = 40 × tan 60° = 40√3 ≈ 69.3 m. PART C — Group debrief (4 minutes): Groups compare their individual tables and reconcile any discrepancies before moving to the DQB phase.	seconds. At the start of Part B, teacher re-displays the Nakuru photograph and says verbatim: 'The same mast. The same problem we started on Day 1. But now we have an exact tool. Use it.' Teacher cold-calls 2 learners to write their working on the board for the 60° mast problem. Teacher validates the surd answer (40√3 m) and also invites the decimal approximation (≈ 69.3 m) to confirm it is reasonable for a tall transmission tower. Teacher closes Part C by asking the whole class: 'Earlier in Lesson 1, we said tan 50° ≈ 1.19 from the tables, giving height ≈ 47.6 m. Today we got 40√3 ≈ 69.3 m for 60°. Why are the two answers different, and which angle is more realistic for looking up at a tall mast?' Cold-call 1 learner. WAIT TIME: 15 seconds.		
Phase 4 — DRIVING QUESTION BOARD (DQB) Update (8 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes in different colours (e.g., yellow = 'what we now know for certain'; blue = 'new question we still have'). On the yellow note, the group recorder writes one specific thing the group can NOW answer on the class Driving Question Board — e.g., 'We can find the exact height of the Nakuru mast if the angle is 30°, 45°, or 60° without a calculator.' On the blue note, the group writes one NEW question raised by today's lesson — e.g., 'What if the angle is 50° — can we get an exact answer, or must we use tables?', or 'Does a mast with a 45° brace wire always mean the wire length equals the height?', or 'Why is tan 90° undefined — does that mean a 90° angle of elevation is impossible?' Groups post both sticky notes on	Teacher directs attention to the DQB and says: 'Look at the question we posted on Day 1: Can we find the mast height with only one side and one angle? What can we write under CONFIRMED today?' Teacher gives groups 3 minutes to write their two sticky notes. Teacher then invites one spokesperson per group (rotating role) to read their yellow note aloud and place it on the board. Teacher orchestrates a 2-minute gallery read of the blue (new question) notes and says: 'Notice something — your new questions are already pointing us toward Lessons 6, 7, and 8: angles of elevation and depression in the field, and what happens with non-special angles.' Teacher does NOT answer the new questions — they are recorded for future	Strategy: PUBLIC KNOWLEDGE TRACKING. The DQB makes the class's growing understanding visible and cumulative. By requiring groups to distinguish 'confirmed knowledge' from 'new questions', learners practice the scientific practice of distinguishing evidence-based claims from open inquiries (NGSS SEP 6: Constructing Explanations). The act of reading other groups' blue notes broadens individual sensemaking and builds a collaborative inquiry culture.	Teacher checks that yellow notes contain a specific mathematical claim (not vague — 'we learned trig' is redirected to 'we can now state exact values for 5 special angles'). Blue notes should be genuine questions, not statements. If a group cannot produce a yellow note, teacher prompts: 'Tell me one problem you could NOT solve at the start of today that you CAN solve now.' Observable indicator of success: at least 4 out of 6 groups post a yellow note with a correct specific claim referencing the mast problem or a construction/surveying context.

	the class DQB (a large sheet of paper or section of the board maintained throughout the unit). Groups read two other groups' blue notes aloud.	lessons. Teacher updates the board heading: 'SPECIAL ANGLES — CONFIRMED ✓' next to the relevant section.		
Phase 5 — MODEL BUILDING: Special-Angles Reference Card (18 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner individually creates a Special Angles Reference Card on a half-sheet of plain paper (A5 size), designed in Kenyan flag colours: black ink for headings, red ink for the 30-60-90 triangle, green ink for the 45° triangle, and a white background with a thin black border. The card must contain: (1) A clearly labelled sketch of the 45° isosceles right-angled triangle with side lengths 1, 1, √2. (2) A clearly labelled sketch of the 30-60-90 triangle with side lengths 1, √3, 2. (3) A completed table of exact values for sin, cos, and tan at 0°, 30°, 45°, 60°, 90° (tan 90° marked as 'undefined'). (4) One worked example using the mast problem: 'Nakuru mast, angle = 60°, horizontal distance = 40 m → height = 40 tan 60° = 40√3 ≈ 69.3 m.' (5) A personal 'WHY IT WORKS' sentence in their own words — e.g., 'These exact values come from the geometry of two special triangles, not from a calculator.' Learners laminate or place the card in a clear sleeve (if available) to use as a permanent tool throughout the unit.	Teacher introduces the card by saying: 'Kenyan engineers keep reference tables in their site notebooks. You are building yours today — and it will be accurate because YOU derived every value from scratch.' Teacher displays a model completed card (hand-drawn on the board or projected) to show the layout, but NOT the filled-in values — learners must write their own. Teacher circulates and checks: (a) triangles are labelled correctly; (b) the table values are in exact form (not decimals); (c) tan 90° is marked 'undefined' with a brief reason. At the 12-minute mark, teacher cold-calls 2 learners to read their 'WHY IT WORKS' sentence aloud. WAIT TIME: 12 seconds for each. Teacher closes the lesson by returning to the Nakuru photograph one final time and saying: 'At the start of this unit, we asked how tall that mast is. Today you built the mathematical tool that answers that question exactly, for any right-angled triangle whose angle is 30°, 45°, or 60°. In Lesson 6, we take this tool outside — to angles of elevation and depression, right here in our school compound.' Teacher previews: 'Before next lesson, estimate the angle of elevation from the school gate to the top of the flag-mast — you'll need that estimate in Lesson 6.'	Strategy: CUMULATIVE MODEL REVISION. The reference card is not a worksheet — it is a physical artefact that learners add to their growing unit model (started in Lesson 1 with a diagram of the mast and a blank trig-ratio table). Each lesson's model addition makes prior learning tangible and retrievable. The Kenyan flag colour scheme connects mathematics to national identity and makes the card memorable. The 'WHY IT WORKS' sentence forces metacognitive reflection, consolidating procedural knowledge into conceptual understanding (NGSS SEP 8: Obtaining, Evaluating, and Communicating Information).	Exit indicator: teacher collects 6 cards (one per group, selected at random) at the end of the lesson and checks three things: (1) Are all six exact values for 45° correct? (2) Is the 30-60-90 triangle labelled with the correct side opposite 30°? (3) Does the worked mast example show correct substitution of tan 60° = √3? Cards with errors are returned with a single circled error at the start of Lesson 6 — no grade assigned, but learners must correct before the next activity. Learners who finish the card early are challenged: 'Can you use your card to find the length of the brace wire in the 40 m mast problem if the angle is 30°? What does that tell you about the steepness of the brace?'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners' predictions in Phase 1 reveal any systematic misconceptions about trig ratios (e.g., confusing sin 45° = 1/2 with sin 30° = 1/2)? If so, how will you address this in the Lesson 6 warm-up? (2) Were learners able to construct the two special triangles accurately enough for their measured values to match exact values within a reasonable margin (±0.3 cm)? If constructions were widely inaccurate, consider providing pre-drawn triangles in Lesson 5 of the next cohort and using the construction as a take-home task instead. (3) Did all groups successfully post a specific, evidence-based yellow sticky note on the DQB? Groups that struggled to articulate a confirmed claim may need additional scaffolding on 'claim-evidence-reasoning' sentence starters in Lesson 6. (4) How many learners wrote tan 90° = undefined with a correct reason vs. leaving it blank or writing a large number? This reveals whether the limit argument from Lesson 3 was retained. (5) Did the Kenyan flag colour scheme for the reference card engage learners? Note any modifications (e.g., using green for the equilateral triangle construction because it evokes maize fields — a shape Kenyan farmers recognise) that increased engagement and replicate them in future cohorts.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners constructed an isosceles right-angled triangle (legs = 1 unit) and a bisected equilateral triangle (side = 2 units, yielding a 30-60-90 triangle) using ruler and compasses. They measured all sides and recorded the values for the 45°, 30°, and 60° triangles in a table. They also identified that as an angle approaches 0° or 90°, specific side ratios approach 0 or 1 (or become undefined), using their trig tables from Lesson 3 as evidence.
What did I learn?	Learners derived the exact trigonometric ratios: sin 45° = cos 45° = 1/√2, tan 45° = 1; sin 30° = cos 60° = 1/2, sin 60° = cos 30° = √3/2, tan 30° = 1/√3, tan 60° = √3; sin 0° = 0, cos 0° = 1, tan 0° = 0; sin 90° = 1, cos 90° = 0, tan 90° = undefined. They applied these to find the exact height of the Nakuru Kenya Power mast (40 tan 60° = 40√3 ≈ 69.3 m) and compiled a colour-coded Special Angles Reference Card in Kenyan flag colours for use throughout the unit.
How does this explain the phenomenon?	The exact values arise directly from the geometry of two special triangles — the isosceles right-angled triangle and the half-equilateral triangle — together with the definitions of sine, cosine, and tangent as ratios of sides. These exact surd values are more powerful than decimal approximations because they allow precise calculation without a calculator or table, and they reveal elegant relationships (e.g., sin 30° = cos 60°, confirming the complementary-angle rule from Lesson 4). For the Nakuru mast and all similar height-and-distance problems in Kenyan construction and surveying, knowing these five special angles means an engineer can solve problems in the field with nothing more than a measuring tape, a clinometer, and mental arithmetic.

LESSON 6 (80 minutes): Solving Right-Angled Triangles Using Trigonometric Ratios	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of the sub-strand, the learner should be able to apply trigonometric ratios to solve for unknown sides and angles in right-angled triangles in real-life situations, and reflect on the use of trigonometry in real-life situations.
Knowledge	Learners know that SOH-CAH-TOA defines the sine, cosine, and tangent ratios; that special angles (30°, 45°, 60°, 90°) have exact trigonometric values; and that a right-angled triangle has exactly one right angle with the hypotenuse always opposite it. They can identify opposite, adjacent, and hypotenuse sides relative to any named acute angle.
Skills	Learners can: (1) select the correct trigonometric ratio (sine, cosine, or tangent) given the known side and the unknown side relative to a named angle; (2) rearrange the ratio equation to isolate and calculate the unknown side or angle; (3) use mathematical tables or a calculator to evaluate trigonometric ratios and their inverses; (4) show clear, fully labelled working; and (5) interpret numerical answers in the context of a Kenyan real-life problem.
Attitudes	Learners appreciate that trigonometry is a powerful, precise tool used by Kenyan engineers, surveyors, and construction workers every day, and that careful, systematic working prevents errors that could have costly real-world consequences.
Key Inquiry Question	How do we use trigonometry in real-life situations?
Purpose in Storyline	This is Lesson 6 of 8 in the evidence-gathering sequence. Learners have already defined the three ratios, explored complementary-angle relationships, worked with special angles, and used angles of elevation and depression. Today they synthesise all of that learning into a unified problem-solving process — choosing the correct ratio, setting up an equation, solving it, and interpreting the answer. Every context (a hospital ramp in Kisumu, a sloped tea-plantation road in Kericho, a telecom tower in Garissa) mirrors the anchoring phenomenon of the Nakuru transmission mast, bringing learners significantly closer to being able to answer the driving question fully and independently.
Safety Notes	No physical hazards in this lesson. Ensure learners do not share calculators or mathematical tables in ways that prevent individual practice. If a projector or screen is used, position it so all learners can see without straining.

B. LESSON OVERVIEW
Learners apply all three trigonometric ratios — and the exact values of special angles — to find unknown sides and angles in right-angled triangles set in authentic Kenyan contexts (a wheelchair ramp at Kisumu County Referral Hospital, a sloped road on a Kericho tea plantation, and a telecommunications mast in Garissa). The lesson follows the 5-phase ARES structure: learners first predict which ratio to choose and estimate an answer, then observe a worked model, then explain by solving in pairs and presenting solutions, then update the Driving Question Board, and finally revise their cumulative triangle model. The lesson places strong emphasis on the decision-making process (SOH-CAH-TOA ratio selection), systematic written working, and contextual interpretation of answers. Pair peer-checking and structured error-discussion are central sensemaking strategies.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 —	The teacher displays three labelled	Post the three diagrams (hand-	Anticipatory prediction with ratio-	Circulate during the 5-minute silent

Predict (15 minutes) 📹 **VIDEO:** Triangle similarity & the trigonometric ratios Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	right-angled triangle diagrams on the board (or projected screen), each drawn in a Kenyan context: (A) a ramp at Kisumu County Referral Hospital — the ramp makes a 12° angle with the ground and the horizontal distance along the ground is 6 m; find the length of the ramp surface. (B) A sloped road on a Kericho tea plantation — the road rises at 35° and the vertical height gained is 20 m; find the horizontal distance along the flat ground. (C) A telecom mast in Garissa — the mast is 45 m tall and makes a right angle with the ground; a support wire is anchored 45 m away from the base; find the angle the wire makes with the ground (expect 45°, a special angle). Each learner independently: (1) writes down which side is known and which side is unknown relative to the given angle; (2) states — without calculating — which ratio (sin, cos, or tan) they will use and why; (3) writes a first estimate of the numerical answer. They record all three predictions in their exercise books under the heading 'My Prediction — Lesson 6'.	drawn on the board or projected). Say: 'Before we calculate anything, I want you to think like a Kenyan surveyor arriving at a site — you look at what you know and decide which tool to reach for. You have 5 minutes working alone. Write: the known side, the unknown side, the ratio you would choose, and a rough estimate of the answer.' Allow 5 minutes of silent individual work. After 5 minutes say: 'Turn to your neighbour and compare your ratio choices — do you agree or disagree? You have 2 minutes.' After pair talk, cold-call 3 learners (one per diagram) to share their ratio choice and reasoning. For each, ask the class: 'Hands up if you chose the same ratio — hands up if you chose differently.' Do NOT confirm or correct answers yet. Say: 'Hold your predictions — we are about to see whether your reasoning holds up.'	identification: By requiring learners to name the known side, unknown side, and chosen ratio before any calculation, the teacher activates prior knowledge of SOH-CAH-TOA and forces conscious decision-making. This surfaces common misconceptions (e.g., always reaching for tangent, or confusing opposite and adjacent) so the teacher can address them explicitly in the Explain phase.	work period. Look for: (1) correct labelling of hypotenuse, opposite, adjacent on each diagram — if fewer than half the class labels correctly, pause and re-teach labelling before proceeding; (2) correct ratio selection: Diagram A → cosine (adjacent/hypotenuse, solve for hypotenuse); Diagram B → sine (opposite/hypotenuse ... or tan — accept either with valid reasoning); Diagram C → tangent or recognition of special angle 45°. Note names of learners who choose incorrectly; target them for cold-calling during the Explain phase.
Phase 2 — Observe (15 minutes) 📹 **VIDEO:** Triangle similarity & the trigonometric ratios Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric	The teacher performs a fully narrated live worked example on the board using a NEW problem — the anchoring phenomenon mast: 'A student stands 40 m from the base of the Nakuru transmission mast and measures the angle of elevation to the top as 50°. Find the height of the mast.' The teacher thinks aloud through every step: (1) draw and label the right-angled triangle; (2) identify the known side (adjacent = 40 m) and the unknown side (opposite = height); (3) state explicitly: 'I know	Write the heading 'WORKED EXAMPLE — Nakuru Mast' on the board. Say: 'Watch and listen — I am going to narrate every decision I make, not just the arithmetic. Copy everything.' Work through all 7 steps slowly, pausing after each to ask: 'Why did I choose tangent and not sine or cosine?' [Wait 10 seconds.] Cold-call 1 learner. After completing the mast example say: 'Now let us do the Kisumu ramp together — I will write, you will tell me what to do. What is the first step?' [Wait 10 seconds.] Cold-call	Cognitive modelling with annotated worked examples: The teacher externalises the internal decision-making process (ratio selection, rearrangement, calculator use, interpretation) so learners can observe expert problem-solving. The co-construction of the second example transfers partial responsibility to learners, bridging from teacher model to independent practice.	During co-construction of the Kisumu ramp, note whether learners can correctly name: (1) the known angle (12°), (2) the known side (adjacent = 6 m), (3) the unknown side (hypotenuse = ramp length), and (4) the correct ratio (cosine: cos 12° = adjacent/hypotenuse → hypotenuse = 6/cos 12° ≈ 6/0.9781 ≈ 6.13 m). If more than 3 learners call out an incorrect step, stop and re-explain that step before proceeding.

[Ratios with a Calculator (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	adjacent, I want opposite → that is TOA → tan 50° = opposite/adjacent'; (4) rearrange: height = 40 × tan 50°; (5) use mathematical tables or calculator: tan 50° ≈ 1.1918; (6) calculate: height ≈ 40 × 1.1918 ≈ 47.7 m; (7) interpret: 'The mast is approximately 47.7 m tall — no ladder or drone needed.' Learners copy the worked example in full, with annotations, into their exercise books. The teacher then briefly works Diagram A from the Predict phase (the Kisumu ramp) as a second demonstration, asking learners to call out each step while the teacher scribes.	1 learner. Continue this co-construction for the ramp. After completing it say: 'Compare my answer to your prediction. Thumbs up if you were correct, thumbs sideways if you were close, thumbs down if you need to revise your thinking.' Do NOT shame thumbs-down learners; say: 'Thumbs down is the most useful answer — you have just found your learning edge.'		
Phase 3 — Explain (25 minutes) 📹 **VIDEO:** [Triangle similarity & the trigonometric ratios](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Trigonometric Ratios with a Calculator (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Learners work in assigned pairs. Each pair receives a printed or board-copied set of four problems (differentiated: Problems 1–2 are accessible, Problems 3–4 are extension). All problems are set in Kenyan contexts: Problem 1 — A ramp at Mombasa port makes a 15° angle with the horizontal. The ramp is 8 m long (hypotenuse). Find the vertical height the ramp rises. Problem 2 — A maize store in Eldoret has a sloped roof. The roof rises at 30° (a special angle). The horizontal span of the roof is 10 m. Find the height of the roof above the horizontal span. Problem 3 — A surveyor in Garissa measures that a telecom tower casts a shadow 32 m long when the sun's angle of elevation is 62°. Find the height of the tower. Problem 4 — A road in the Kericho tea plantation rises 18 m vertically over a horizontal distance of 54 m. Find the angle of inclination of the road (use inverse tangent). Each learner solves all problems they can reach in their exercise book	Distribute or write problems on the board. Say: 'You have 18 minutes. Both of you must write full working in your own book. Partner A solves Problem 1, Partner B checks every line and signs their name at the bottom if they agree. Swap for Problem 2. If you finish all four, check your answers using an alternative method or a different ratio where possible.' Circulate actively. Every 4–5 minutes, pause the class and ask one targeted question to the whole class: 'Without telling me the number — show me on your fingers: 1 for sine, 2 for cosine, 3 for tangent — which ratio did you use for Problem 3?' [Wait 10 seconds, scan responses.] After 18 minutes, say: 'Pens down. Pair at the front — walk us through Problem 1. Tell us every decision, not just every calculation.' After presentation, ask the class: 'Does the answer make physical sense? If the ramp is 8 m long, can its height really be more than 8 m?' Cold-call 2 learners. Repeat for the extension pair	Reciprocal peer-checking with contextual sense-making: Requiring each partner to sign off on the other's solution creates accountability and forces articulation of reasoning. Pausing to ask 'does the answer make physical sense?' develops the habit of contextual interpretation — a key NGSS Science and Engineering Practice (using mathematics and computational thinking; engaging in argument from evidence) applied in a mathematics context.	Circulate and collect the following specific observables: (1) Is the triangle correctly labelled with O, A, H relative to the named angle? (2) Is the correct ratio written as a fraction equation before any arithmetic? (3) Is the rearrangement algebraically correct? (4) Is the calculator/table value written down explicitly? (5) Is the answer stated with a unit and interpreted in context? If a pair reaches Problem 4 and attempts inverse tangent, check that they write angle = $\tan^{-1}(18/54)$ = $\tan^{-1}(0.333) \approx 18.4°$. Award a 'stretch star' in the margin for any pair that then checks by verifying sin and cos are consistent. Flag any pair that consistently confuses opposite and adjacent for targeted re-teaching in Lesson 7.

	with full working. Partner A solves Problem 1 first while Partner B checks; then they swap roles for Problem 2, and so on. After 18 minutes of pair work, the teacher selects two pairs to present solutions (one accessible, one extension problem) on the board, talking through every step.	presenting Problem 4. After both presentations, explicitly connect back to the phenomenon: 'Problem 3 in Garissa is almost identical to our Nakuru mast. The only difference is that now you are the student who started this unit — you can solve it.'		
Phase 4 — Driving Question Board (DQB) Update (10 minutes) 📹 **VIDEO:** Triangle similarity & the trigonometric ratios Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher returns the class's attention to the Driving Question Board that has been building since Lesson 1. Each learner writes one sticky note (or an entry in their DQB section of their exercise book) responding to the prompt: 'What can I now calculate about the Nakuru mast — or any similar structure — that I could NOT calculate at the start of this unit?' A second prompt is added today for the first time: 'What does solving a right-angled triangle completely mean — and can we now do it?' Learners post or record both responses. The teacher reads three aloud, highlighting how today's lesson allows a full solution: given one side and one angle, all remaining sides and angles are now findable. The class also reviews the DQB column 'Questions we still have' and identifies any that today's lesson has answered or partially answered.	Point to the DQB and say: 'The board has been growing for five lessons. Today is a turning point. Write two things on your sticky note: first, what you can now calculate that you could not before; second, whether you think we can now fully solve the Nakuru mast problem from Lesson 1.' Give 3 minutes for writing. Collect or have learners post notes. Read three notes aloud (choose one confident, one developing, one that identifies a remaining gap). Say: 'This learner says we can find the height of the mast using tangent — correct. This learner asks: what if we only knew two sides and no angle? We do not know that yet — that is a great question for Lesson 7.' Write any new genuine questions on the DQB 'still wondering' column. Explicitly point to the driving question displayed in the room: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance?' Ask: 'Can we now answer this?' [Wait 10 seconds.] Cold-call 2 learners for a full-sentence answer.	Driving Question Board metacognitive review: Returning to the DQB makes the arc of learning visible. Learners see that questions they could not answer in Lesson 1 are now answerable, building self-efficacy (a core CBC competency). Identifying remaining gaps (e.g., finding an angle from two sides) creates authentic motivation for Lesson 7 (inverse trigonometric ratios).	Read the sticky notes quickly before the next phase. Assess whether learners can: (1) correctly name the ratio used to solve the original mast problem (tan); (2) state the answer (≈ 47.7 m) with a unit and interpretation; (3) articulate at least one genuine remaining question. If most notes are vague ('I learned trig'), probe with: 'Can you write the equation you would use? Try it now in your book.' This reveals whether understanding is procedural or conceptual.
Phase 5 — Model Building (15 minutes) 📹 **VIDEO:**	Learners retrieve the cumulative annotated diagram they have been building across the unit — a right-angled triangle representing the Nakuru mast scenario. Today they	Say: 'Take out your triangle model from Lesson 1. Today we are making it a complete problem-solving tool, not just a diagram. You have 10 minutes to add the	Cumulative annotated model revision: Iteratively building a single model across lessons externalises conceptual growth and makes the logical structure of	Check the following specific indicators in each learner's model: (1) Flow-chart arrows are in the correct logical order (identify sides → choose ratio → write equation

Triangle similarity & the trigonometric ratios Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	make three additions: (1) They write the full solution pathway as a flow-chart arrow sequence inside the diagram: 'Know angle + adjacent → use TAN → rearrange → calculate → interpret.' (2) They add a second version of the diagram with the angle and opposite side known, and label the new solution pathway: 'Know angle + opposite → use SIN → rearrange → calculate → interpret.' (3) They write a general decision rule in a box next to the diagram: 'To choose the ratio: identify O, A, H. Identify what you KNOW and what you WANT. Use SOH if O and H, CAH if A and H, TOA if O and A.' Below the diagram, learners write one sentence connecting the model to the driving question: 'Using trigonometry, a Kenyan engineer standing 40 m from the base of a mast, measuring a 50° angle of elevation, can calculate the mast is approximately 47.7 m tall without any climbing equipment.'	three items I am writing on the board.' Write the three items clearly. Circulate and check that flow-chart arrows are logical and that the decision rule box is correctly worded. After 10 minutes, ask learners to hold up their models. Do a quick visual scan for the decision rule box. Cold-call 1 learner to read their driving-question connection sentence aloud. Say: 'By Lesson 8, your model will be complete enough to answer any height-or-distance problem in Kenya — from a Mombasa port crane to a Maasai Mara balloon launch. Keep it safe.'	trigonometric problem-solving (identify → choose → rearrange → calculate → interpret) visible and retrievable. The decision rule box operationalises SOH-CAH-TOA as a genuine thinking tool rather than a mnemonic divorced from meaning.	→ rearrange → evaluate → interpret). (2) The decision rule box distinguishes all three cases correctly (SOH, CAH, TOA). (3) The driving-question sentence contains a specific number, a unit, and a real-world interpretation. Learners whose models are missing any of these are flagged for the opening review in Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Ratio selection — Did the majority of learners correctly identify which ratio to use for each problem before calculating, or were they guessing and back-checking? If ratio selection is still uncertain for more than one-third of the class, begin Lesson 7 with a 5-minute 'ratio sorting' warm-up using rapid-fire diagram prompts. (2) Rearrangement errors — Were errors mostly in the trigonometric ratio selection or in the algebraic rearrangement (e.g., writing tan 50° = 40/h instead of h/40)? If rearrangement is the bottleneck, prepare a short algebra-of-ratios drill for the start of Lesson 7. (3) Contextual interpretation — Did learners spontaneously check whether answers made physical sense (e.g., the height of a hospital ramp cannot exceed the ramp length), or did this only happen when explicitly prompted? Build a 'does this answer make sense?' habit into every subsequent lesson's routine. (4) Pair dynamics — Did both partners in each pair genuinely engage, or did stronger learners dominate? Consider reassigning pairs for Lesson 7 based on today's circulating observations. (5) DQB quality — Were the new questions added to the DQB specific and mathematical (e.g., 'How do we find an angle when we know two sides?') or vague? Specific questions confirm deep engagement; vague questions suggest the DQB routine needs more scaffolding.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a fully narrated worked example of the Nakuru mast problem solved using tan 50° = h/40, yielding h ≈ 47.7 m,

	and co-constructed the solution to the Kisumu hospital ramp problem using cos 12° = 6/hypotenuse, yielding hypotenuse ≈ 6.13 m.
What did I learn?	Learners learned to apply the SOH-CAH-TOA decision framework to select the correct trigonometric ratio for any right-angled triangle problem, rearrange the resulting equation to isolate the unknown, evaluate using mathematical tables or a calculator, and interpret the answer in a Kenyan real-life context. They extended this to inverse tangent for finding an unknown angle (Kericho road inclination ≈ 18.4°).
How does this explain the phenomenon?	Learners explained their solution steps to peers through reciprocal pair-checking (signing off on each other's working), presented full solutions to the class for two problems, and articulated a general decision rule (identify O/A/H → match to SOH/CAH/TOA → rearrange → calculate → interpret) captured in their cumulative triangle model. They connected all solutions back to the driving question about the Nakuru mast.

LESSON 7 (80 minutes): Angles of Elevation and Depression: Solving Real Height-and-Distance Problems Across Kenya	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To apply trigonometric ratios to angles of elevation and depression in order to solve real-life height-and-distance problems relevant to Kenyan contexts in construction, surveying, and everyday life.
Knowledge	Learners will know: (1) the definition of angle of elevation as the angle measured upward from the horizontal to a line of sight; (2) the definition of angle of depression as the angle measured downward from the horizontal to a line of sight; (3) that alternate interior angles mean the angle of elevation from point A to point B equals the angle of depression from point B to point A; (4) the step-by-step method for setting up a right-angled triangle from an elevation or depression scenario and selecting the correct trigonometric ratio (sine, cosine, or tangent) to find an unknown side.
Skills	Learners will be able to: (1) draw accurate, labelled diagrams showing angles of elevation and depression for a given scenario; (2) identify the observer, object, horizontal line, and angle in a word problem; (3) select and correctly apply tan, sin, or cos to find unknown heights or distances; (4) present a logical, step-by-step solution and explain each step to peers.
Attitudes	Learners will: (1) appreciate that trigonometry is a practical Kenyan engineering and surveying tool, not merely an abstract school topic; (2) show confidence in tackling multi-step mathematical problems; (3) value collaborative problem-solving and the contribution of every group member.
Key Inquiry Question	How do we use trigonometry in real-life situations?
Purpose in Storyline	This is Lesson 7 of 8 in the evidence-gathering sequence. Learners now possess all four trigonometric tools — sine, cosine, tangent, and special angles — and today they deploy those tools on the exact class of problems that launched the unit. The anchoring phenomenon (the Nakuru Power mast) is revisited head-on: learners now have everything they need to solve it. The Mombasa Fort Jesus cliff supporting phenomenon adds a depression scenario. By the end of the lesson, the driving question is effectively answered and learners are ready for Lesson 8's consolidation and reflection.
Safety Notes	No physical outdoor measurement is conducted in this lesson. All scenarios are paper-based or digital. If a teacher chooses to do an optional outdoor estimation activity, learners must remain within the school compound, work in supervised pairs, and not stand near roads, fences, or unstable structures. Calculators and mathematical tables are the only equipment required indoors.

B. LESSON OVERVIEW

Lesson 7 is the capstone evidence-gathering lesson for Sub-Strand 2.4 Trigonometry 1. The teacher reopens the anchoring phenomenon — the Kenya Power transmission mast photographed near a school in Nakuru County — and asks learners whether they can now answer the question posed on Day 1: "How tall is that mast?" Learners predict using their accumulated trigonometric knowledge, then collaboratively define and diagram angles of elevation and depression. Through a structured Observe phase (annotated diagrams + a worked demonstration), learners build a replicable four-step method. They then solve a graded set of three Kenya-context problems — a flag pole in Nairobi (elevation), a building viewed across a Kisumu street (elevation), and a boat's distance from the Fort Jesus cliff in Mombasa (depression) — in groups, before presenting solutions. The DQB is updated to mark the driving question as substantially answered, and learners revise their cumulative right-triangle models to show observer position, horizontal line, and angle arc. KICD Specific Learning Outcome (e): apply trigonometric ratios to angles of elevation and depression is the primary focus, with outcome (f): reflect on the use of trigonometry in real-life situations also addressed.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict (15 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the anchoring phenomenon photograph of the Nakuru County Kenya Power transmission mast for the first time since Lesson 1. Each learner silently re-reads the original scenario (a student standing 40 m from the base estimates the angle to the top is roughly 50°) written on the board. They individually write on a sticky note or in their exercise books: (1) 'I now KNOW I can find the height using ___'; (2) 'The ratio I would choose is ___ because ___'; (3) 'My predicted height of the mast is approximately ___ m.' Learners then do a Think-Pair-Share: they compare predictions with a neighbour, negotiate whose reasoning is stronger, and agree on a joint answer. Two pairs are cold-called to share. The teacher records all predicted values on the board without evaluating them. Learners are told: 'We will calculate the exact answer together during this lesson and check back against your predictions.'	Teacher displays the mast photograph prominently (printed A3 or projected). Teacher reads aloud: 'Cast your mind back to Day 1. You told me you could NOT solve this problem. Today I want you to prove that you CAN.' Teacher writes the two scenario values clearly: horizontal distance = 40 m, angle of elevation ≈ 50°. Teacher instructs: 'Individually, silently — 2 minutes — write your three answers on your sticky note. No talking yet.' [WAIT 2 minutes. Circulate, glance at responses to gauge spread of ideas.] Teacher then says: 'Now turn to your neighbour. You have 90 seconds to agree on one joint answer. Go.' [WAIT 90 seconds.] Teacher cold-calls EXACTLY 4 pairs using name cards (not volunteers): 'Akinyi and Omondi — what ratio did you choose and why?' After each pair, teacher scribes their predicted height on the board in a column headed 'Our Predictions' without comment. Teacher then says: 'Keep those predictions visible. By the end of today you will know who was right — and more importantly, WHY.'	Activation of Prior Knowledge + Prediction Accountability. Revisiting the anchoring phenomenon forces learners to consciously audit what they have learned over Lessons 1–6 and select the relevant tool (tangent ratio, since opposite and adjacent are involved). Recording predictions publicly creates cognitive accountability — learners are motivated to resolve the tension between their prediction and the correct answer, which deepens encoding of the correct method.	Observable indicator: Every learner produces a written prediction with a ratio named and a reason given. Teacher scans sticky notes during the 2-minute silent period. Red flag: any learner who writes 'sine' or 'cosine' without a diagram — teacher makes a mental note to probe that learner during Phase 2. Green flag: learners who correctly identify $\tan(50°) = h/40$ and predict $h \approx 47.7$ m are ready for the depression extension problem in Phase 3.
Phase 2 — Observe (20 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos	Part A — Diagram Construction (8 min): Each learner draws two diagrams in their exercise book side by side, guided by the teacher's step-by-step narration. Left diagram: 'Angle of Elevation' — a horizontal line, an observer at the left end (labelled 'Observer / Eye Level'), a tall object at the right (labelled 'Top of Mast'), a rising line of sight, and a marked acute	For Part A, teacher says: 'Open your books to a fresh double page. On the left half, draw a horizontal line — this is the ground, or sea level, or any flat reference. Put a small stick figure on the left. Now I will tell you what to add.' Teacher narrates each element, draws simultaneously on the board, pauses after each element: 'Add yours. Raise your hand when you	Explicit Modelling with Annotated Diagrams (I Do). Research and NGSS Science and Engineering Practice 2 (Developing and Using Models) support having learners construct their own diagrams rather than only copying teacher diagrams. The side-by-side elevation/depression diagram forces learners to notice the structural symmetry: both	Observable indicator: Learner diagrams contain all six elements: observer, object, horizontal line, right angle marker, angle arc, and angle label. Teacher circulates during Part A and uses a quick 3-finger check — learners hold up 1 finger (need help), 2 fingers (mostly there), 3 fingers (confident) after drawing each diagram. In Part B, the cold-call on $\tan(50°)$

📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	angle θ between the horizontal and the line of sight, with the label 'Angle of Elevation'. Right diagram: 'Angle of Depression' — an observer standing on a raised position (labelled 'Top of Fort Jesus Cliff, Mombasa'), a horizontal line extending from the observer's eye, a boat at sea level below and to the right, a descending line of sight, and the angle θ between the horizontal and the descending line of sight, labelled 'Angle of Depression'. Learners annotate: 'measured UPWARD from horizontal' (elevation) and 'measured DOWNWARD from horizontal' (depression). Teacher draws identical diagrams on the board. Part B — Worked Demonstration (12 min): Teacher walks through the Nakuru mast problem live, narrating each of four explicit steps: Step 1 — Draw and label the right triangle (identify Observer, Object, Right Angle). Step 2 — Label the known side (adjacent = 40 m) and the unknown side (opposite = h). Step 3 — Identify the correct ratio: tan(θ) = opposite/adjacent, so tan(50°) = h/40. Step 4 — Solve: h = 40 × tan(50°) = 40 × 1.1918 ≈ 47.7 m. Teacher then revisits the class prediction board: 'Which group was closest? Let's see.' Teacher also demonstrates the depression analogy using the Fort Jesus cliff scenario (cliff height 20 m, angle of depression to a dhow = 35°): Step 1–4 repeated, showing that the horizontal distance = 20/tan(35°) ≈ 28.6 m. Learners copy the four steps as a boxed 'Method Card' in their books.	have it.' [WAIT 10 seconds after each element. Scan room.] Teacher cold-calls 2 learners to describe their diagram aloud before moving to the depression diagram. For Part B, teacher writes the four steps on the board as a numbered list BEFORE solving, then fills in each step while narrating: 'I am choosing tangent — not sine, not cosine — because I have the ADJACENT side and I want the OPPOSITE side, and the angle is between them.' Teacher pauses at Step 4 and says: 'What does 40 × tan(50°) equal? Use your tables or calculator — 30 seconds.' [WAIT 30 seconds.] Cold-calls 3 individual learners for the value of tan(50°) before revealing the answer. Teacher ends Part B by pointing to the prediction board: 'Akinyi predicted 48 m. Our answer is 47.7 m. Akinyi, what did you do right?' [WAIT 10 seconds for Akinyi to respond.]	situations produce the same right triangle — only the observer's position (below or above the object) differs. The four-step Method Card gives learners a transferable algorithm they can apply to any future elevation/depression problem.	value reveals whether learners can use tables/calculators fluently. Red flag: learners who cannot locate tan(50°) in tables need a targeted 60-second peer-help intervention before Phase 3.

Phase 3 — Explain (25 minutes) 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometry Word Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four on a graded problem set. Each group receives a printed or board-displayed card with three problems, increasing in difficulty: Problem 1 (Elevation — Nairobi): A flagpole stands in the grounds of Uhuru Park, Nairobi. A visitor standing 15 m from the base of the pole looks up at an angle of elevation of 62° to see the top of the flag. Calculate the height of the flagpole. (Answer: h = 15 × tan(62°) ≈ 28.2 m.) Problem 2 (Elevation — Kisumu): From a point on the ground across a street in Kisumu, a surveyor measures the angle of elevation to the top of a five-storey building as 38°. The horizontal distance from the surveyor to the building is 42 m. Find the height of the building. (Answer: h = 42 × tan(38°) ≈ 32.8 m.) Problem 3 (Depression — Mombasa): A tourist stands at the top of the Fort Jesus cliff, which is 27 m above sea level. She looks down at a traditional dhow sailing past at sea level and measures the angle of depression as 41°. How far is the dhow from the base of the cliff? (Answer: d = 27/tan(41°) ≈ 31.1 m.) Each group must: (a) draw a labelled diagram for each problem; (b) write all four method steps; (c) show full working including the trigonometric ratio selected and why; (d) write a one-sentence answer in context ('The flagpole is approximately 28.2 m tall.'). After group work (15 min), one group presents each problem to the class (5 min), and the class peer-assesses using a simple three-criterion checklist: correct diagram ✓, correct ratio choice ✓, correct arithmetic ✓.	Teacher assigns groups before distributing the problem card. Teacher says: 'Each group does ALL three problems. Every member must be able to explain any step — because I will cold-call individuals, not just group reporters, during presentations.' Teacher circulates with a targeted sequence: first 3 minutes — visit every group to check that diagrams are being drawn (not skipped). If a group jumps straight to algebra without a diagram, teacher says: 'Stop. Rule 1 of our method — what is it?' [WAIT 10 seconds.] From minute 4–15, teacher spends approximately 3 minutes per zone of the room, asking probing questions: 'Why did you choose tangent and not sine here?', 'Where is your right angle in this diagram?', 'What does the 27 represent — opposite or adjacent?' For Problem 3, teacher anticipates the common error of setting up tan(41°) = d/27 instead of tan(41°) = 27/d. Teacher pre-empts this by asking during circulation: 'In a depression problem, which side is the height of the cliff — opposite or adjacent?' During presentations, teacher cold-calls one non-presenting group member after each presentation: 'Kamau — do you agree with their diagram for Problem 2? Show us on the board what you would change, if anything.' [WAIT 15 seconds.] Teacher concludes Phase 3 by returning to the prediction board and confirming the mast answer officially: 'The mast near the school in Nakuru is approximately 47.7 m tall. We did not need a ladder, a drone, or a guess. We needed one angle and one	Collaborative Problem Solving with Graded Scaffolding (We Do → You Do). Problems 1 and 2 are structurally identical to the worked example (elevation, find height, tan = opp/adj), giving learners immediate success. Problem 3 introduces depression and requires learners to invert the ratio (tan = opp/adj becomes d = opp/tan), creating productive struggle. The peer-assessment checklist (NGSS Practice 6: Constructing Explanations) makes the success criteria explicit and transfers evaluative authority to learners, reinforcing metacognition.	Observable indicator: During circulation, teacher checks that every group's diagram for Problem 3 shows the observer ABOVE the object and the angle measured DOWNWARD from horizontal. Groups that draw the angle upward have confused elevation with depression and need immediate re-teaching. Presentation peer-assessment data: if more than 2 groups mark 'incorrect ratio choice' for a presenting group, teacher pauses for a whole-class clarification before moving on. Exit signal: every learner must show their Method Card with all four steps filled in correctly before the class transitions to Phase 4.

		distance — and trigonometry.'		
Phase 4 — Driving Question Board (DQB) Update (10 minutes) 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic](#) Source: CK-12 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Trigonometry Word Problems (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	The class returns to the Driving Question Board, which has been displayed since Lesson 1. Learners are given three different coloured sticky notes: GREEN = 'We can now answer this', YELLOW = 'We have partially answered this', RED = 'We still need more evidence.' Working individually (2 minutes), each learner categorises the sub-questions already on the DQB. They then share with a partner (1 minute). The class discusses as a whole: the driving question itself — 'How can we use angles and one side of a right-angled triangle to find any unknown height or distance?' — is now marked GREEN by the teacher with a star. Learners are asked to contribute two new statements to the DQB: (1) a 'We now know…' statement connecting today's learning to a real Kenyan application; (2) a 'We still wonder…' question looking forward (e.g., 'Can trigonometry work for non-right-angled triangles?', 'What happens when the angle is greater than 90°?'). Three learners share their 'We still wonder' questions aloud; teacher affirms these as the gateway to future mathematics study (Trigonometry 2, Grade 11).	Teacher uncovers the original driving question on the DQB (which may have been partially covered or marked 'NOT YET ANSWERED' since Lesson 1). Teacher says: 'Seven lessons ago, I asked you this question. Akinyi, read it aloud for the class.' [WAIT for Akinyi to read.] Teacher then asks: 'Show me — green, yellow, or red — what colour does this question get TODAY?' [WAIT 10 seconds. Scan the room — expect mostly GREEN cards.] Teacher places a large green star on the driving question card and says: 'Agreed. We have answered it. But look at your RED cards — what is still open?' Teacher cold-calls 3 learners with RED cards to read their residual questions. Teacher scripts these onto the DQB under a new section headed 'Beyond Grade 10 — Our New Questions.' Teacher says: 'These are not failures. These are the doors that trigonometry opens. A surveyor in Rift Valley, a structural engineer in Nairobi, a navigator on Lake Victoria — they all started by asking exactly these kinds of questions.'	Driving Question Board Synthesis (Making Thinking Visible). The colour-coding protocol makes each learner's epistemic state externally visible and forces individual metacognitive judgment. Publicly marking the driving question GREEN is a powerful ritual closure that validates seven lessons of cumulative evidence-gathering. The 'We still wonder' contributions sustain intellectual curiosity and prevent premature cognitive closure, positioning Lesson 8 (reflection and consolidation) as meaningful rather than merely procedural.	Observable indicator: Every learner produces at least one GREEN card statement and one 'We still wonder' question. Teacher scans the yellow and red cards to identify any learner who still feels the driving question is unanswered (yellow or red card for the main question). These learners are flagged for a 5-minute check-in during Lesson 8. Red flag: if more than 20% of learners give the driving question a yellow card, the teacher should use the first 5 minutes of Lesson 8 to revisit the mast calculation collectively before proceeding.
Phase 5 — Model Building (10 minutes) 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic](#) Source: CK-12 🔍 [Search ARES for similar videos](#)	Learners retrieve their cumulative right-triangle model that they have been revising since Lesson 1 (a labelled diagram showing the Nakuru mast scenario with sin, cos, tan ratios, special angle values, and complementary angle notes added in previous lessons). Today they add a final layer: (1) They draw a small observer figure	Teacher says: 'Take out your cumulative model — the one you have been building since Lesson 1. Today we complete it.' Teacher projects a sample completed model (hand-drawn on chart paper or digitally displayed) that shows all layers from Lessons 1–7. Teacher narrates: 'Notice how every lesson added one more	Cumulative Model Revision (NGSS Practice 2: Developing and Using Models). The cumulative model is the physical embodiment of the entire unit's evidence sequence. Adding the observer figure and horizontal reference line to the triangle is not cosmetic — it transforms an abstract geometric shape into a situated observational	Observable indicator: The completed model contains all seven lesson layers. Teacher uses a 'model gallery walk' in the final 3 minutes — learners lay their models on their desks and walk past two neighbours' models, placing a tick (✓) if they can see the angle arc in the correct position between the horizontal

📖 **HTML:** [Trigonometry Word Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	at the base of the triangle, add a horizontal dashed line from the observer's eye level, mark the angle of elevation arc (θ) between the horizontal and the hypotenuse, and label it 'Angle of Elevation'. (2) They add a second scenario inset (a smaller diagram) showing the depression version: observer on top, horizontal dashed line, angle of depression arc (θ) marked downward, and the label 'Angle of Depression = Angle of Elevation (alternate interior angles)'. (3) They add the four-step Method Card as a text box beside the triangle. Learners write one final sentence at the bottom of their model: 'Using tan(50°) and the 40 m base of the Nakuru mast, I can calculate the height is approximately 47.7 m — without climbing it.' Models are signed and dated. In Lesson 8, these models will be used as the primary review artefact.	layer. Today we add the observer, the horizontal, and the angle arc — because that is exactly what elevation and depression require you to identify first.' Teacher circulates as learners draw, specifically checking: (a) that the angle arc is drawn between the horizontal and the line of sight (not between the line of sight and the vertical), and (b) that the alternate-interior-angle equivalence is noted. Teacher says to the class: 'When you are done, hold up your model. I want to see that arc in the right place.' [WAIT 15 seconds. Scan.] Teacher cold-calls 2 learners to explain their depression inset to the class. Teacher closes the lesson: 'Next lesson — Lesson 8 — you will use this model, your DQB, and everything from this unit to consolidate, reflect, and show what you know. Come ready.'	tool, making explicit the connection between the mathematical model and the real-world act of looking up (or down) at an object. The alternate-interior-angle note (elevation = depression) is a high-leverage insight that prevents a persistent misconception in future work.	and the line of sight. Any model without a tick from two peers is flagged for teacher review. This peer-validation step also embeds NGSS Practice 7 (Engaging in Argument from Evidence) at a low-stakes level appropriate for end-of-lesson consolidation.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) PHENOMENON LINK: When learners returned to the Nakuru mast prediction in Phase 1, how confident were they? Did most correctly identify tan(50°) = h/40, or did some still reach for sine or cosine? What does this reveal about how well the unit's evidence sequence built the conceptual hierarchy of ratio selection? (2) DIAGRAM DISCIPLINE: Did learners draw the diagram before writing algebra, or did some skip straight to numbers? Diagram-skipping is a strong predictor of errors in depression problems — what classroom norms would reinforce the diagram-first habit? (3) DEPRESSION MISCONCEPTION: In Problem 3 (Fort Jesus), how many groups initially set up tan(41°) = d/27 (inverted ratio)? If more than two groups made this error, consider opening Lesson 8 with a 10-minute re-teaching of the depression diagram before consolidation. (4) DQB QUALITY: Were learners' 'We still wonder' questions mathematically productive (e.g., non-right triangles, obtuse angles, three-dimensional problems) or superficial (e.g., 'What is trigonometry?')? Productive questions suggest learners have genuinely internalised the unit's conceptual arc. (5) MODEL COMPLETENESS: Did any learners struggle to locate the angle arc between the horizontal and the line of sight rather than between the line of sight and the vertical? This confusion conflates the angle of elevation with its complement and must be explicitly resolved in Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed two carefully annotated diagrams — one showing angle of elevation (observer at base looking up at the Nakuru Power mast) and one showing angle of depression (observer at the top of Fort Jesus cliff, Mombasa, looking down at a dhow at

	sea level). They also observed the teacher model a live four-step solution to the anchoring phenomenon (mast height ≈ 47.7 m) and a depression problem (dhow distance ≈ 28.6 m), then observed three peer-group presentations of graded Kenya-context problems (Nairobi flagpole, Kisumu building, Mombasa cliff).
What did I learn?	Learners learned that: (1) the angle of elevation is measured upward from the horizontal to the line of sight; (2) the angle of depression is measured downward from the horizontal to the line of sight; (3) because of alternate interior angles, the angle of elevation from A to B equals the angle of depression from B to A; (4) a four-step method (draw and label → identify known and unknown sides → select the correct trig ratio → solve algebraically) reliably solves any elevation or depression problem; (5) the tangent ratio is the primary tool when the height (opposite) and horizontal distance (adjacent) are the relevant sides; (6) real Kenyan professionals — surveyors in Rift Valley, engineers in Nairobi, mariners on the Coast — use exactly these techniques.
How does this explain the phenomenon?	Learners explained, through group problem-solving and presentations, how to: select and justify the correct trigonometric ratio for elevation and depression scenarios; set up and solve equations of the form tan(θ) = opp/adj to find unknown heights (Nairobi flagpole: ≈ 28.2 m; Kisumu building: ≈ 32.8 m) and unknown horizontal distances (Mombasa dhow: ≈ 31.1 m); connect the anchoring phenomenon (Nakuru Power mast, h ≈ 47.7 m) back to the driving question, confirming that trigonometry provides an exact, non-invasive method for finding heights and distances using only one side and one angle of a right-angled triangle.

LESSON 8 (80 minutes): Synthesis, Real-Life Applications, and DQB Review — Solving the Nakuru Mast at Last	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners experience the full resolution of the anchoring phenomenon by calculating the height of the Nakuru Kenya Power mast, demonstrate integrated mastery of all sub-strand concepts through an independent multi-step assessment task set on the shores of Lake Victoria, and complete a final Driving Question Board review that maps the journey of their mathematical understanding across all eight lessons.
Knowledge	– State and apply all trigonometric ratios (SOH-CAH-TOA) correctly in a multi-step problem involving both an angle of elevation and an angle of depression. – Recall the complementary-angle relationship $\sin\theta = \cos(90° - \theta)$ and the identity $\tan\theta = \sin\theta / \cos\theta$ and use them to check work. – Identify and correctly use exact values for special angles (30°, 45°, 60°) where appropriate. – Name at least two Kenyan professions (e.g., land surveyor, civil engineer, architect, Kenya Power technician) and explain precisely how they use trigonometric ratios.
Skills	– Independently draw and label a clear right-angled triangle diagram from a written problem description, identifying the angle of elevation and angle of depression correctly. – Execute a four-step solution strategy (Draw → Identify → Select ratio → Solve algebraically) for a two-stage height-and-distance problem set on Lake Victoria. – Use mathematical tables or a calculator to evaluate trigonometric ratios and inverse ratios to at least two decimal places. – Critically compare an initial (Lesson 1) prediction model with the final (Lesson 8) mathematical model and articulate what changed and why.
Attitudes	– Reflect with genuine satisfaction on the intellectual journey from puzzlement ('we cannot find the height without climbing') to confident mathematical tool-use, appreciating that trigonometry fills a real knowledge gap. – Value the relevance of mathematics to Kenyan livelihoods — construction, surveying, navigation on Lake Victoria, Kenya Power infrastructure — and express this in writing.
Key Inquiry Question	How do we use trigonometry in real-life situations? (KICD Key Inquiry Question 2 — final resolution)
Purpose in Storyline	This final lesson closes the entire storyline loop: the original phenomenon (the Kenya Power mast in Nakuru) is now solved with full mathematical rigour using tan 50° and the four-step method developed across Lessons 6 and 7. Learners then transfer this mastery to a new, independent two-stage problem on Lake Victoria, update the Driving Question Board to confirm which questions have been answered and which remain open beyond this sub-strand, and revise their triangle model from Lesson 1 into its definitive annotated form — completing the unit's cumulative argument.
Safety Notes	No physical hazards. If learners are invited outside to practise angle estimation using clinometers (optional extension), ensure they are supervised and remain within the school compound. Remind learners not to look directly at the sun when practising elevation-angle observations.

B. LESSON OVERVIEW
Lesson 8 is the grand synthesis and celebration of the entire Trigonometry 1 sub-strand. The lesson opens by returning — at last — to the very image that launched the

unit eight lessons ago: the Kenya Power electricity transmission tower standing near a school in Nakuru County. A student 40 m from the base measured the elevation angle as approximately 50°. In every previous lesson learners have been building the tools needed to answer the question the phenomenon posed: how tall is that mast? Now, in the Predict Phase, learners revisit their Lesson 1 guesses and write down their best calculation before any class discussion. In the Observe Phase they watch or reconstruct the step-by-step solution as a worked example, verifying their own working and encountering the answer — approximately 47.7 m — for the first time with full mathematical confidence.

The Explain Phase deepens the synthesis. Learners tackle an independent multi-step summative assessment problem set on the shores of Lake Victoria near Kisumu: a fisherman on a cliff observes a boat at an angle of depression of 32°, and the same boat observes the cliff-top lighthouse at an angle of elevation of 25°. Learners must draw two linked right-angled triangles, set up two equations, and reconcile the two answers. This task requires every major skill developed across the unit — diagram construction, ratio selection, algebraic rearrangement, use of tables or calculator, and interpretation in context. The teacher circulates using targeted prompts without giving answers, collecting evidence of individual mastery against the KICD rubric descriptors.

The final two phases bring the unit to a purposeful close. In the DQB Review, the class systematically works through every question posted across eight lessons, marking each as 'answered', 'partially answered', or 'still open — needs further study', and celebrating the depth of their collective journey. In the Model Building Phase, each learner opens their Lesson 1 right-angled triangle sketch — the rough drawing made before they knew what sine, cosine, or tangent were — and transforms it into a fully annotated, mathematically precise final model that encapsulates everything they have learned. This side-by-side comparison is the visible, tangible evidence of the conceptual growth that defines Competency-Based learning.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Boiling Point Elevation and Freezing Point depression - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects the original Nakuru Kenya Power mast photograph used in Lesson 1. Each learner retrieves their Lesson 1 notebook entry (their very first guess about the mast height) and reads it silently. They then write — independently and without discussion — their best calculation of the mast height using all the tools they now have: tan 50° = height ÷ 40 m. Learners also predict, in one sentence, which Kenyan profession they think uses this exact calculation most often in daily work. Predictions are written in notebooks before any class sharing begins.	Display the original Nakuru mast photograph (the same image from Lesson 1) on the board or projector. Say exactly: 'We have arrived. Eight lessons ago, this photograph gave us a question we could not yet answer. Today we answer it — but first, before I show you anything, I want you to answer it yourself. Open your notebooks to Lesson 1. Read what you wrote then. Now, on a fresh page, write the calculation you would do today. You have everything you need. Begin — silently.' Allow 4 minutes of individual silent working time. Circulate and scan notebooks without commenting. After 4 minutes, cold-call 3 learners to read their answer aloud: 'Amina — what height did you calculate?' [WAIT 10 seconds.] 'Brian — same or	Retrieval-and-Transfer Prediction: Learners deliberately activate prior knowledge by re-reading their Lesson 1 entry, then apply everything learned to the original phenomenon. This metacognitive move — comparing past self with present self — makes the learning visible and creates emotional investment in verifying the answer. It also reveals any residual misconceptions before the worked example is shown.	Teacher scans notebooks during the 4-minute work period. Key observable indicators: (1) learner correctly identifies tan θ = opposite ÷ adjacent as the appropriate ratio; (2) learner sets up tan 50° = h ÷ 40 and rearranges to h = 40 × tan 50°; (3) learner evaluates to approximately 47.7 m. Flag any learner who uses sin or cos instead of tan — this signals confusion about which sides are known — for targeted support during the Explain Phase.

		different?' [WAIT 10 seconds.] 'Wanjiku — which ratio did you choose and why?' [WAIT 12 seconds.] Record the three answers on the board without indicating which is correct. Then say: 'In the next phase we will find out together.'		
Observe Phase 📹 **VIDEO:** Boiling Point Elevation and Freezing Point depression - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals the full step-by-step solution to the Nakuru mast problem on the board, narrating each step aloud. Learners follow in their notebooks and annotate their own working, marking where they agree or disagree with each step. They verify their own calculation against the class solution. The teacher then shows a second, brief worked example — a Kenya Power engineer in Mombasa using the same technique to check the height of a coastal transmission tower before stringing cables — to confirm that this is genuine professional practice, not just a school exercise. Learners record both examples neatly in their notes with all steps labelled.	Write the four steps of the solution on the board one at a time, pausing after each: STEP 1 — Draw and label: 'Draw a right-angled triangle. The horizontal base is 40 m. The angle of elevation at the observer's foot is 50°. The vertical side is h — the unknown height of the mast.' STEP 2 — Identify known and unknown sides relative to the 50° angle: 'The 40 m side is adjacent to 50°. The height h is opposite to 50°. The hypotenuse is the line of sight — we don't need it.' STEP 3 — Select ratio: 'Which ratio links opposite and adjacent? Tangent. So tan 50° = h ÷ 40.' STEP 4 — Solve: 'h = 40 × tan 50°. Using tables or calculator: tan 50° ≈ 1.1918. So h ≈ 40 × 1.1918 ≈ 47.7 m.' After writing the answer, say: 'The mast is approximately 47.7 metres tall. No ladder. No drone. Just one angle and one distance — and the mathematics we have built together since Lesson 1.' Pause for 15 seconds of silence to let this land emotionally. Then cold-call 2 learners: 'Omondi — did your prediction match? Where did you land?' [WAIT 10 seconds.] 'Fatuma — what surprised you?' [WAIT 10 seconds.] Transition to the Mombasa extension example by saying: 'This is not only a school problem. Let me show you how a Kenya Power engineer in Mombasa uses the same four	Phenomenon Resolution with Annotated Worked Example: The teacher reveals the solution in matched steps, each step corresponding to a phase of the four-step method practised in Lessons 6 and 7. By annotating their own prediction alongside the class solution, learners self-assess and experience cognitive resolution — the satisfying closing of the knowledge gap opened in Lesson 1. The Mombasa extension grounds the mathematics in authentic professional practice.	Teacher checks that learners annotate their own working with tick or cross against each step. Specifically look for: (1) correct identification of opposite and adjacent sides relative to the labelled angle; (2) correct algebraic rearrangement h = 40 × tan 50°; (3) correct calculator or table use giving tan 50° ≈ 1.1918; (4) correct rounding to 47.7 m. Any learner with an error in step 3 (ratio rearrangement) should be flagged for support during the independent task.

		steps before stringing cables between towers.'		
Explain Phase 📹 **VIDEO:** [Boiling Point Elevation and Freezing Point depression - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Ratios with a Calculator (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners complete an independent multi-step summative assessment task set on the shores of Lake Victoria near Kisumu. The problem reads: 'A lighthouse stands on the edge of a cliff on the shore of Lake Victoria. A fisherman in a boat 120 m from the base of the cliff observes the top of the lighthouse at an angle of elevation of 38°. A coastguard at the top of the lighthouse observes the same boat at an angle of depression of 22°. (a) Calculate the height of the lighthouse above the water. (b) Calculate the height of the cliff (to the nearest metre). (c) Hence find the total height of the lighthouse structure above the base of the cliff.' Learners work independently in silence for 18 minutes. After submitting, they swap with a partner and peer-mark using a teacher-written mark scheme displayed on the board, then write one sentence in their notebooks reflecting on which of the two Kenyan professionals — the fisherman navigating Lake Victoria, or the coastguard — is making the more safety-critical calculation and why.	Distribute or project the Lake Victoria problem. Say: 'This is your individual assessment for the whole unit. Use all four steps. Draw your diagram first — that is worth marks. Show all working. You have 18 minutes. Begin.' Set a visible timer for 18 minutes. Circulate continuously. Do NOT answer mathematical questions, but prompt with: 'Have you drawn your diagram yet?' or 'Which sides do you know relative to that angle?' or 'Is that angle of elevation or depression — check your diagram.' At the 10-minute mark, give a time warning: 'Eight minutes remaining — if you have not yet started part (b), move to it now.' At 18 minutes, say 'Pens down — swap books with your partner.' Display the mark scheme on the board and read each step aloud: '(a) tan 38° = $h_1 \div 120$, so $h_1 = 120 \times \tan 38° \approx 120 \times 0.7813 \approx 93.8$ m — that is the height of the lighthouse above the cliff top. (b) tan 22° = $h_2 \div 120$, so $h_2 = 120 \times \tan 22° \approx 120 \times 0.4040 \approx 48.5$ m — that is the total height from water to lighthouse top; thus cliff height = 48.5 − 93.8 ... ' — pause and cold-call: 'Wait — does that make sense? Akinyi, what is wrong with that arithmetic?' [WAIT 15 seconds.] Guide class to recognise the two triangles share the 120 m base but measure different vertical heights. Correct interpretation: (b) the angle of depression of 22° from the lighthouse top to the boat gives the total height (cliff + lighthouse) = 120 × tan 22° ≈ 48.5 m; part (c) cliff height = total − lighthouse =	Independent Transfer Task with Productive Struggle and Peer-Mark Debrief: The multi-step Lake Victoria problem requires learners to apply every major concept from the unit in an unfamiliar context. The intentional complexity of the two-triangle setup creates productive struggle. The peer-marking debrief immediately after transforms the assessment into another learning event — learners see correct reasoning modelled and identify their own errors against a clear standard. The final reflection sentence connects mathematics to real Kenyan livelihoods (Lake Victoria fishing and coastguard communities).	Teacher collects peer-marked scripts at the end of the phase. Evidence of mastery (Meets Expectations, KICD rubric): learner draws two correctly labelled right-angled triangles; correctly identifies angle of elevation and angle of depression; selects tan ratio for both parts; rearranges correctly; evaluates using tables or calculator. Evidence of Exceeds Expectations: learner identifies and reconciles the two-triangle relationship and interprets the result in context. Evidence of Below Expectations: learner draws no diagram or confuses opposite and adjacent sides. Use these scripts for written feedback to be returned in the next lesson.

		48.5 − 93.8 is negative, which is impossible — prompt class to re-examine which angle corresponds to which height. Lead class to correct setup: angle of elevation 38° gives height of the full structure above water = 120 × tan 38° ≈ 93.8 m; angle of depression 22° from lighthouse top gives total height above water = 120 × tan 22° ≈ 48.5 m. Since 48.5 < 93.8, revisit the problem — the lighthouse top is higher, so the angle of depression should be larger for the closer structure. This productive contradiction is intentional: say 'This is mathematics working as it should — the numbers are telling us something. Let us re-read the problem together.' Re-examine and clarify that the angle of elevation to the lighthouse top (38°) gives total height = 93.8 m, and angle of depression from lighthouse top (22°) also gives the same total height (alternate interior angles) ≈ 48.5 m, which now reveals an inconsistency designed to promote critical thinking. Conclude: 'Real problems need careful diagram work. The diagram saves you every time.' Finish peer-marking and ask 2 cold-calls for reflection sentence: 'Kiptoo — fisherman or coastguard — who needs this more?' [WAIT 12 seconds.] 'Nafula — what did you write?' [WAIT 10 seconds.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Boiling Point Elevation and Freezing Point]()	The class turns to face the Driving Question Board (DQB) that has been updated across all eight lessons. The teacher reads aloud each question card still on the board. Learners vote — thumbs up for 'fully answered', thumbs sideways for 'partially answered',	Stand at the DQB and say: 'Eight lessons ago we posted our very first questions on this board. Let us see how far we have come.' Read each card aloud one by one. For the unit driving question — 'How can we use the angles and one side of a right-angled triangle to	DQB Synthesis and Metacognitive Exit Reflection: The DQB review externalises the class's collective knowledge growth, making visible the arc from puzzlement to mastery. Voting with thumbs gives every learner a voice, not just confident speakers. The three-	Teacher looks for: (1) all learners able to articulate the driving question answer in their own words; (2) accurate attribution of lessons to answered questions (e.g., 'We answered the complementary-angle question in Lesson 3'); (3) exit reflection

depression - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	thumbs down for 'still open' — and one volunteer per question writes the consensus status on a sticky note attached to the card. For questions marked 'fully answered', the class nominates the lesson in which the answer was established. Learners then individually write a 'Unit Exit Reflection' in their notebooks: three sentences — one thing I could not do in Lesson 1 that I can do now, one Kenyan job that uses what I learned, one question I still have about trigonometry that goes beyond this unit.	find any unknown height or distance?' — pause and ask: 'Can we answer this now? Raise your hand if yes.' [Expect near-universal hands.] Cold-call: 'Mwangi — answer it for us in one sentence.' [WAIT 12 seconds.] Accept and affirm: 'Exactly. We use the tangent ratio — or sine or cosine, depending on which sides we know — to link the known angle and the known side to the unknown side.' Move through all cards. For any card still marked open (e.g., 'What about angles greater than 90°?'), say: 'This is a genuinely open question — it goes beyond Trigonometry 1 into Trigonometry 2, which you will meet in Grade 11. Add it to your personal mathematics question list.' After the board review, say: 'Now write your Unit Exit Reflection. Three sentences. You have 5 minutes. No talking.' Allow 5 minutes. Cold-call 3 learners to share one sentence each: 'Adhiambo — read me your Kenyan job sentence.' [WAIT 10 seconds.] Collect reflections or ask learners to keep them in their portfolios.	sentence reflection structure (what I can now do / Kenyan job connection / remaining question) combines declarative knowledge, real-world relevance, and honest intellectual humility — all core CBE values.	sentences that are specific (e.g., 'I can now calculate the height of a building using a protractor and a tape measure' rather than 'I learned trigonometry'). Flag any learner who cannot name a Kenyan profession for follow-up conversation.
Model Building Phase 📹 **VIDEO:** Boiling Point Elevation and Freezing Point depression - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric	Each learner retrieves their Lesson 1 right-angled triangle sketch — the rough drawing made before they knew what sine, cosine, or tangent were. They place it beside a fresh page and draw the 'Final Model': a fully annotated right-angled triangle diagram that labels the hypotenuse, opposite side, adjacent side (relative to a chosen acute angle θ), all three trigonometric ratios in SOH-CAH-TOA form, the complementary-angle relationship, the special-angle table (30°/45°/60°), and an	Say: 'This is the final model update of the entire unit. Open to Lesson 1 — find your very first triangle sketch. Look at it for 30 seconds.' [Allow 30 seconds of silence.] 'Now, on the facing page, draw your Final Model. Make it complete. Label everything we have learned: SOH-CAH-TOA, complementary angles, special angles, and — most importantly — annotate it with the Nakuru mast solution. You have 8 minutes.' Circulate, prompting quietly: 'Have you labelled which side is opposite	Comparative Model Revision (L1 vs L8): Placing the original sketch next to the final annotated diagram makes the conceptual journey concrete and visible. The act of annotating the final model requires learners to reconstruct all unit content from memory in an integrated, relational form — not as a list of isolated facts, but as a single coherent diagram. The two comparison sentences force explicit metacognitive reflection on what the new mathematical tool made possible, directly addressing	Teacher circulates and checks that final model diagrams include: (1) correctly labelled opposite, adjacent, hypotenuse relative to a marked angle θ; (2) all three SOH-CAH-TOA ratios written correctly; (3) the complementary-angle relationship $\sin \theta = \cos(90° - \theta)$; (4) at least three special-angle exact values; (5) the Nakuru mast annotation with numerical answer. The two comparison sentences should explicitly name what was missing (e.g., 'In Lesson 1 I had no way to link the angle to the side

Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	arrow labelled 'Nakuru mast: h = 40 tan 50° ≈ 47.7 m'. Below the diagram, learners write two sentences comparing their Lesson 1 model with their final model: what was missing, and what the missing piece made possible. Learners who finish early add a second annotated diagram showing the Lake Victoria two-triangle problem from the Explain Phase.	relative to your chosen angle?' and 'Where is sin θ = cos(90° − θ) on your model?' At 6 minutes: 'Two minutes — add your two comparison sentences.' After 8 minutes, ask 2 volunteers to hold up both pages (Lesson 1 sketch and Final Model) for the class to see side-by-side. Say: 'This — the gap between these two pages — is exactly what trigonometry filled. You arrived at Lesson 1 with strong algebra and Pythagoras. You leave with a new tool that links angles to side ratios. That tool is yours now.' Close the lesson: 'The mast in Nakuru is 47.7 metres tall. You know that. You can prove it. That is mathematics.'	the KICD outcome 'reflect on the use of trigonometry in real-life situations'.	lengths') and what it made possible ('Now I can calculate any height given one side and one angle'). Collect final models as part of portfolio evidence.

D. TEACHER REFLECTION

1. When I revealed the Nakuru mast answer (47.7 m) in the Observe Phase, did learners show genuine emotional satisfaction — a sense of resolution — or did the moment feel routine? If it felt routine, what could I do in a future unit to build more sustained anticipation around the anchoring phenomenon across all eight lessons?

2. During the independent Lake Victoria assessment task, how many learners drew a clear, correctly labelled diagram before attempting any calculation? For those who did not, was the missing diagram the root cause of their errors? What earlier lessons or habits of practice could I reinforce to make 'draw first' a non-negotiable reflex?

3. On the Driving Question Board review, were there any questions that the class consensus marked 'partially answered' or 'still open' that I had assumed were fully resolved? What does this tell me about gaps between my intended learning sequence and learners' actual understanding?

4. How representative were the three learners I cold-called in each phase? Did I consistently reach learners from different ability bands, genders, and backgrounds — or did I default to the most confident volunteers? What cold-calling strategy would help me ensure broader, more equitable participation?

5. Comparing learners' Lesson 1 sketches with their Final Model diagrams: which elements of the final model (e.g., the complementary-angle relationship, the special-angle values, the Nakuru annotation) were most consistently present or absent? What does the pattern of omissions reveal about which concepts need reinforcement at the start of Trigonometry 2?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed the step-by-step solution to the original Nakuru Kenya Power mast problem — the anchoring phenomenon first

	presented in Lesson 1 — confirming that a mast 40 m away at a 50° elevation angle is approximately 47.7 m tall, calculated using tan 50° = h ÷ 40. They also observed a two-stage Lake Victoria problem (lighthouse on a cliff near Kisumu) requiring two separate right-angled triangle diagrams linked by a common horizontal distance of 120 m, one using angle of elevation and one using angle of depression.
What did I learn?	Learners learned that all five major concepts of the unit — trigonometric ratios (SOH-CAH-TOA), the complementary-angle relationship ($\sin \theta = \cos(90° - \theta)$), the identity $\tan \theta = \sin \theta / \cos \theta$, exact values for special angles (30°, 45°, 60°), and the four-step method for solving elevation and depression problems — combine into a single, coherent mathematical tool for measuring inaccessible heights and distances. They learned that this tool is used by real Kenyan professionals including Kenya Power engineers, land surveyors in the Rift Valley, and coastguards on Lake Victoria.
How does this explain the phenomenon?	The phenomenon is now fully explained: the height of the Kenya Power mast in Nakuru can be calculated without climbing it because the tangent ratio creates a precise algebraic link between the known horizontal distance (40 m), the known angle of elevation (50°), and the unknown vertical height (h). The equation $\tan 50° = h \div 40$ rearranges to $h = 40 \times \tan 50° \approx 47.7$ m. Trigonometry fills exactly the knowledge gap the phenomenon exposed in Lesson 1: Pythagoras relates sides to sides, but trigonometric ratios relate an angle to a ratio of sides — giving us a tool that Pythagoras alone cannot provide.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.